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Gout

[Clinical Policy Bulletins](#) | [Medical Clinical Policy Bulletins](#)

Number: 0810

Table Of Contents

[Policy](#)

[Applicable CPT / HCPCS / ICD-10 Codes](#)

[Background](#)

[References](#)

Policy History

[Last Review](#) 

03/27/2026

Effective: 12/10/2010

Next Review: 01/28/2027

[Review History](#) 

[Definitions](#) 

Policy

Scope of Policy

This Clinical Policy Bulletin addresses gout for commercial medical plans.

For Medicare criteria, see [Medicare Part B Criteria \(https://www.aetna.com/health-care-professionals/medicare/part-b-step.html\)](https://www.aetna.com/health-care-professionals/medicare/part-b-step.html).

Note: Requires Precertification:

Precertification of pegloticase (Krystexxa) is required of all Aetna participating providers and members in applicable plan designs. For precertification of pegloticase (Krystexxa) call (866) 752-7021 or fax (888) 267-3277. For Statement of Medical Necessity (SMN) precertification forms, see [Specialty Pharmacy Precertification \(https://www.aetna.com/health-care-professionals/health-care-professional-forms.html\)](https://www.aetna.com/health-care-professionals/health-care-professional-forms.html).

Note: Site of Care Utilization Management Policy applies for pegloticase (Krystexxa) beginning January 1, 2026. For information on site of service for pegloticase (Krystexxa), see [Utilization Management Policy on Site of Care for Specialty Drug Infusions \(https://www.aetna.com/health-care-professionals/utilization-management/drug-infusion-site-of-care-policy.html\)](https://www.aetna.com/health-care-professionals/utilization-management/drug-infusion-site-of-care-policy.html).

Additional Information

[Clinical Policy Bulletin](#)

[Notes](#) 

I. Medical Necessity

Aetna considers the following as medically necessary for the management of gout:

A. Diagnostic Testing:

1. Measurement of blood uric acid levels
2. Measurement of erythrocyte sedimentation rate
3. Polarized light microscopy for identification of crystal in synovial fluids obtained from joints or bursas (as well as material aspirated from tophaceous deposits, if any)
4. Magnetic resonance imaging for gouty tophus, which may mimic an infectious or neoplastic process;

B. Treatment:

1. Pegloticase (Krystexxa)

a. Criteria for Initial Approval

For members with a diagnosis of chronic gout when *all* of the following criteria are met:

- i. Member is 18 years of age or older; *and*
- ii. The requested medication will not be used concomitantly with oral urate-lowering therapies; *and*
- iii. The member has at least 2 gout flares per year that were inadequately controlled by colchicine or non-steroidal anti-inflammatory drugs (NSAIDs), or at least 1 gout tophus or gouty arthritis; *and*
- iv. Member has had an inadequate response to or a clinical reason for not completing at least a three-month trial (see [Appendix A](#)) with the following medications at the medically appropriate maximum doses:
 - a. Allopurinol or febuxostat
 - b. Probenecid (alone or in combination with allopurinol or febuxostat); *and*
- v. The member meets one of the following criteria:
 - a. The requested medication will be co-administered with weekly oral methotrexate and folic acid or folinic acid supplementation; *or*
 - b. The member has a contraindication to or clinical reason to avoid oral methotrexate therapy (see [Appendix B](#)).

Aetna considers pegloticase therapy as experimental, investigational, or unproven for all other indications.

b. Continuation of Therapy

For treatment of chronic gout when *all* of the following criteria are met:

- i. Member is 18 years of age or older; *and*
- ii. The requested medication will not be used concomitantly with oral urate-lowering therapies; *and*
- iii. The member meets one of the following:
 - a. The requested medication will be co-administered with weekly oral methotrexate and folic acid or folinic acid supplementation; *or*
 - b. The member has a contraindication to or clinical reason to avoid oral methotrexate therapy (see [Appendix B](#)); *and*
- iv. Member has not had two consecutive uric acid levels above 6 mg/dL since starting treatment with the requested medication; *and*
- v. Member is experiencing benefit from therapy (e.g., serum uric acid levels less than 6 mg/dL, reduction of tophi, reduction of symptoms and/or flares).

2. Canakinumab (Ilaris)

For management of gout and pseudogout flares see [CPB 0881 - Canakinumab \(Ilaris\) \(0881.html\)](#).

II. Experimental, Investigational, or Unproven

The following are considered experimental, investigational, or unproven for the management of gout because the effectiveness of these approaches has not been established (not an all-inclusive list):

A. Testing:

- Baseline gut microbiome as a predictive biomarker of response to probiotic adjuvant treatment in the management of gout
- Digital tomosynthesis
- Dual-energy computed tomography-based radiomics for

differentiating individuals with and without gout flares

- Evaluation of gut microbiota
- Genetic testing (for the management of gout)
- Indolelactic acid as a metabolic biomarker for the diagnosis of gout
- Measurement of 24-hour urine uric acid levels
- Measurement of blood lead levels
- Measurement of microRNAs
- Measurement of salivary uric acid levels
- Measurement of scalp hair uric acid levels
- Measurement of serum cystatin C level (as a marker of the renal function damage and inflammation in persons with gout)
- Measurement of serum ionized magnesium as an index of bone erosion in gouty arthritis
- Measurement of synovial fluid uric acid level
- Next-generation sequencing profiling of mitochondrial genomes
- Plasma profiling of amino acids (for differential diagnosis of acute gout from asymptomatic hyperuricemia)
- Use of metabolites (e.g., adenosine, DL-2-aminoadipic acid, guanosine, hypoxanthine, kynurenic acid, leukotriene B4, xanthine) for early detection of gout;

B. Treatment/Prevention:

- Nanoparticles for treatment of gouty arthritis
- Pricking-blood therapy for the treatment of acute gouty arthritis
- Use of sodium-glucose transport protein 2 (SGLT2) inhibitors for the prevention of incident gout in individuals with type-2 diabetes mellitus
- Pegloticase for all other indications not listed in Section I.B., including the following (not an all-inclusive list):
 - Asymptomatic hyperuricemia
 - Persons with glucose-6-phosphate dehydrogenase (G6PD) deficiency.

III. Related Policies

- [CPB 0300 - Hair Analysis \(../300_399/0300.html\)](#)
- [CPB 0881 - Canakinumab \(Ilaris\) \(../800_899/0881.html\)](#)

Dosage and Administration

Krystexxa is a clear and colorless solution, containing 8 mg of pegloticase expressed as concentrations of uricase protein. It is available in two forms: an 8 mg/50 mL (0.16 mg/mL) Ready-to-Use single-dose vial and an 8 mg/mL To-be-Diluted single-dose vial, both intended for intravenous infusion.

The recommended dosage is Krystexxa 8 mg given as an intravenous infusion every two weeks, co-administered with weekly oral methotrexate 15 mg and folic acid or folinic acid supplementation. Krystexxa alone may be used in persons for whom methotrexate is contraindicated or not clinically appropriate.

Source: Horizon Therapeutics USA, 2025

CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description
CPT codes covered if selection criteria are met:	
70540 - 70543, 71550 - 71552, 73218 - 73223, 73718 - 73723	Magnetic resonance imaging [gouty tophi only]

Code	Code Description
84550	Uric acid, blood
85651	Sedimentation rate, erythrocyte; non-automated
85652	automated
CPT codes not covered for indications listed in the CPB:	
<i>Digital tomosynthesis for the diagnosis of gout, pricking blood therapy, evaluation of gut microbiota, Baseline gut microbiome, Dual-energy computed tomography-based radiomics - no specific code</i>	
81460	Whole mitochondrial genome (eg, Leigh syndrome, mitochondrial encephalomyopathy, lactic acidosis, and stroke-like episodes [MELAS], myoclonic epilepsy with ragged-red fibers [MERFF], neuropathy, ataxia, and retinitis pigmentosa [NARP], Leber hereditary optic neuropathy [LHON]), genomic sequence, must include sequence analysis of entire mitochondrial genome with heteroplasmy detection [next generation sequencing profiling mitochondrial genomes]
81465	Whole mitochondrial genome large deletion analysis panel (eg, Kearns-Sayre syndrome, chronic progressive external ophthalmoplegia), including heteroplasmy detection, if performed [next generation sequencing profiling mitochondrial genomes]
82136	Amino acids, 2 to 5 amino acids, quantitative, each specimen
82139	Amino acids, 6 or more amino acids, quantitative, each specimen
82610	Cystatin C
83520	Immunoassay for analyte other than infectious agent antibody or infectious agent antigen; quantitative, not otherwise specified [measurement of microRNA for the diagnosis of gout]
83655	Lead
84560	Uric acid; other source
83497	Hydroxyindolacetic acid, 5-(HIAA)
83735	Magnesium
Other CPT codes related to the CPB:	
82960	screen

Code	Code Description
96365	Intravenous infusion, for therapy, prophylaxis, or diagnosis (specify substance or drug); initial, up to 1 hour
+96366	each additional hour (List separately in addition to code for primary procedure)
+96367	additional sequential infusion of a new drug/substance, up to 1 hour (List separately in addition to code for primary procedure)
+96368	concurrent infusion (List separately in addition to code for primary procedure)
96372	Therapeutic, prophylactic, or diagnostic injection (specify substance or drug); subcutaneous or intramuscular [canakinumab, rilonacept]
96379	Unlisted therapeutic, prophylactic, or diagnostic intravenous or intra-arterial injection or infusion
HCPCS codes covered if selection criteria are met:	
J0638	Injection, canakinumab, 1 mg
J2507	Injection, Pegloticase, 1 mg
HCPCS codes not covered for indications listed in the CPB:	
<i>Sodium- glucose transport protein (SGLT2) inhibitors (Canagliflozin, Dapagliflozin, Empagliflozin and Ertugliflozin), Nanoparticles – no specific code</i>	
Other HCPCS codes related to the CPB:	
J1808	Injection, folic acid, 0.1 mg
J8610	Methotrexate; oral, 2.5 mg
J8611	Methotrexate (jylamvo), oral, 2.5 mg
J8612	Methotrexate (xatmep), oral, 2.5 mg
ICD-10 codes covered if selection criteria are met:	
M1A.00X0 - M1A.9XX1	Gout
M11.20 - M11.29	Other chondrocalcinosis [pseudogout flares]
ICD-10 not covered for indications listed in the CPB:	
E79.0	Hyperuricemia without signs of inflammatory arthritis and tophaceous disease [asymptomatic]

Background

Gout is a crystalline arthropathy predominantly observed in adult men.

Plasma uric acid concentrations in excess of 6 to 7 mg/dl lead to the formulation of monosodium urate crystals, which in turn are deposited in joints and tissues. Symptoms of gout include recurrent inflammatory arthritis that can lead to permanent joint destruction; the development of tophi which can be painful when inflamed and limit joint mobility, and uric acid urolithiasis. In the United States, there is a self-reported prevalence of 1.4% in men and 0.6% in women. The prevalence increases with age and has been estimated to reach 9% in men and 6% in women over 80 years of age. The overall prevalence has been increasing over time.

Other treatment options include NSAIDs, colchicine, probenecid and allopurinol. Uloric and allopurinol exhibit a similar mechanism of action; however, Uloric may be more efficacious in patients with mild to moderate renal failure.

Gout is a condition caused by the over-production or under-excretion of uric acid, resulting in the deposition of monosodium urate crystals in the joints or soft tissue. The disease is often, but not always, associated with increased blood uric acid levels. The four phases of gout are (i) asymptomatic hyperuricemia, (ii) acute gouty arthritis, (iii) inter-critical gout, and (iv) chronic tophaceous gout. The peak incidence of gout occurs in patients 30 to 50 years old, and the condition is much more common in men than in women. Individuals with asymptomatic hyperuricemia do not require specific treatment; however, attempts should be made to decrease their urate levels by encouraging them to make dietary and lifestyle modifications (e.g., a low carbohydrate, high protein and unsaturated fat diet). Acute gout most commonly affects the first metatarsal joint of the foot, but the small joints of the hands, wrists and elbows may also be involved. Gout rarely occurs in the shoulders, hips, sacroiliac joints or spine. Gout in the elderly differs from classical gout found in middle-aged men in several respects: it has a more equal gender distribution, frequent polyarticular presentation with involvement

of the joints of the upper extremities, fewer acute gouty episodes, a more indolent chronic clinical course, and an increased incidence of tophi, which are deposits of monosodium urate crystals in people with longstanding high levels of uric acid in the blood and are commonly seen in conjunction with gout. Long-term diuretic use in patients with hypertension or congestive cardiac failure, renal insufficiency, prophylactic low-dose aspirin, and alcohol abuse (particularly by men) are factors associated with the development of hyperuricemia and gout in the elderly (Pittman and Bross, 1999; Harris et al, 1999; Agudelo and Wise, 2000; Agudelo and Wise, 2001).

Segal and Albert (1999) stated that diagnosis of the crystal-induced arthritides is primarily based on microscopic identification of crystals in synovial fluid. Harris and colleagues (1999) noted that definitive diagnosis requires joint aspiration with demonstration of birefringent crystals in the synovial fluid under a polarized light microscope.

While blood level of uric acid has been commonly used as a diagnostic indicator of hyperuricemia and gout, the value of salivary level, scalp hair level, as well as 24-hour urine level of uric acid in diagnosing gout has not been established. Microscopic analysis by means of compensated polarized light and culture of synovial fluid helps differentiate gouty arthritis from other arthropathies, and the presence of monosodium urate crystals establishes the diagnosis of gout. When gout is suspected, yet the initial examination does not reveal the telltale crystals, re-examination of synovial fluid is warranted. It is important to note that diagnosis of gout does not rule out the possibility of concurrent arthritic conditions (Uy et al, 1996; Owen-Smith et al, 1998; Kobayashi et al, 1998; Pittman and Bross, 1999; Schlesinger et al, 1999).

Report of a task force of the Standing Committee for International Clinical Studies Including Therapeutics on the diagnosis of gout (Zhang et al, 2006a) stated that radiographs have little role in diagnosis, though in late or severe gout radiographical changes of asymmetrical swelling and subcortical cysts without erosion may be useful to differentiate chronic gout from other joint conditions.

Treatment goals include termination of the acute attack, prevention of recurrent attacks and prevention of complications associated with the

deposition of urate crystals in tissues. Pharmacotherapy remains the mainstay of treatment. Acute attacks may be terminated with the use of non-steroidal anti-inflammatory drugs (NSAIDs), colchicine or intra-articular injections of corticosteroids. Probenecid, sulfinpyrazone and allopurinol can be used to prevent recurrent attacks. In patients with peptic ulcer disease, selective cyclo-oxygenase-2 (COX-2) inhibitors provide another treatment option. In the presence of renal impairment, allopurinol is the treatment of choice for urate-lowering therapy, but doses of allopurinol and colchicine must be adjusted. Urate-lowering therapy should only be used if recurrent episodes of gout occur despite aggressive attempts to reverse or control the underlying causes. It should not be introduced or discontinued during an acute episode of gout. Obesity, alcohol consumption and certain foods and medications can contribute to hyperuricemia. These risk factors should be identified and modified (Pittman and Bross, 1999; McGill, 2000; van Doornum and Ryan, 2000; Zhang et al, 2006b).

Caution should be exercised when prescribing NSAIDs for the treatment of acute gouty arthritis in the elderly. Short-acting NSAIDs (e.g., diclofenac and ketoprofen) are preferred, but these drugs are not recommended in patients with peptic ulcer disease, renal failure, uncontrolled hypertension or cardiac failure. Colchicine is poorly tolerated in the elderly and is best avoided. Intra-articular and systemic corticosteroids are increasingly being used for treating acute gouty flares in elderly patients with medical disorders contraindicating NSAID therapy. Urate-lowering drugs are poorly tolerated and the frequent presence of renal impairment in the elderly renders these drugs ineffective. Allopurinol is the urate-lowering drug of choice, but its use in the elderly is associated with an increased incidence of both cutaneous and severe hypersensitivity reactions. To minimize this risk, the dosage of allopurinol must be kept low (Fam, 1998).

Cronstein and Terkeltaub (2006) stated that despite the detailed mechanistic picture for gouty inflammation, there are no placebo-controlled, randomized clinical studies for any of the therapies commonly used, although comparative studies have demonstrated that many NSAIDs are equivalent to indomethacin with respect to controlling acute gouty attacks. In general, the 1st-line of anti-inflammatory therapy for acute gout is NSAIDs, and the selective COX-2 inhibitor, celecoxib, can

be used where appropriate. The 2nd-line of treatment is glucocorticoids, given systemically (intramuscular, intravenous, or oral) or intra-articularly. Alternatively, synthetic adrenocorticotrophic hormone is effective, partly via induction of adrenal glucocorticosteroids and partly via rapid peripheral suppression of leukocyte activation by melatonin receptor 3 signaling. The 3rd-line of treatment is oral colchicine, which is highly effective when given early in an acute gouty attack, but it is poorly tolerated because of predictable gastrointestinal side effects.

The task force of the Standing Committee for International Clinical Studies Including Therapeutics on the management of gout (Zhang et al, 2006b) noted that recommended drugs for acute gout attacks were oral NSAIDs, oral colchicine, or joint aspiration and injection of corticosteroid. Urate-lowering therapy is indicated in patients with recurrent acute attacks, arthropathy, tophi, or radiographical changes of gout. Allopurinol was confirmed as effective long-term urate-lowering therapy. If allopurinol toxicity occurs, options include other xanthine oxidase inhibitors, allopurinol de-sensitization, or a uricosuric. The uricosuric benzbromarone is more effective than allopurinol and can be used in patients with mild-to-moderate renal insufficiency but may be hepatotoxic. When gout is associated with the use of diuretics, the diuretic should be stopped if possible. For prophylaxis against acute attacks, either colchicine 0.5 to 1 mg daily or an NSAID (with gastro-protection if indicated) is recommended.

The clinical guideline on the management of initial gout in adults by the University of Texas at Austin (2009) included pharmacotherapies (e.g., colchicine, corticosteroids [intra-articular or systemic], NSAIDs, and vitamin C), as well as non-pharmacological management (e.g., avoidance of heat therapy, co-morbidity management, diet including coffee [2 cups of coffee daily], low alcohol diet, low-fat dairy diet, low fructose diet [especially avoiding sugar-sweetened soft drinks], and low purine diet [avoidance of red meats, seafood], ice therapy, and rest of affected joint).

In April 2009, the U.S. Food and Drug Administration (FDA) approved febuxostat (Uloric), a non-purine analog xanthine oxidase inhibitor and is the first new urate-lowering gout drug in more than 40 years. In August 2009, the FDA approved colchicine (Condydon) for the treatment of acute gout. Several other pharmaceutical companies are also conducting

clinical trials to test new drugs for the treatment of acute and chronic gout; one of them is pegloticase, a pegylated recombinant uricase that converts urate into the easily excretable allantoin (Schlesinger 2010).

Yue and associates (2008) described the pharmacokinetics and pharmacodynamics of pegloticase in 40 gout patients. Pegloticase was administered as intravenous infusions every 2 weeks at 4- and 8-mg doses, or every 4 weeks at 8- or 12-mg doses for 12 weeks. Serum pegloticase concentrations, plasma urate, and serum antibody response were determined. Population pharmacokinetics and pharmacodynamics analyses were performed. Data were modeled simultaneously, and co-variables were examined (age, antibody response, body weight, gender, ideal body weight, and race). The dosing regimens to maintain uric acid levels below the therapeutic target of 6 mg/dL were then predicted by the model. The pharmacokinetics were best described by a 1-compartment linear model, while the pharmacodynamics model was fitted as a direct effect of pegloticase on uric acid concentrations with a suppressive maximum effect attributed to drug (E_{max}) function. Pegloticase suppressed uric acid levels up to 83 %. Weight only affected clearance and volume of distribution. No co-variables affected pharmacodynamics. Simulation suggests pegloticase administered at 8 mg every 2 or 4 weeks as 2-hour intravenous infusions will maintain uric acid levels well under 6 mg/dL.

In a phase II, randomized study, Sundy et al (2008) evaluated the effectiveness of pegloticase in achieving and maintaining plasma urate levels of less than 6 mg/dl in gout patients in whom other treatments have failed, and assessed the pharmacokinetics and safety of pegloticase. A total of 41 patients were randomized to undergo 12 to 14 weeks of treatment with pegloticase at 1 of 4 dosage levels: (i) 4 mg every 2 weeks, (ii) 8 mg every 2 weeks, (iii) 8 mg every 4 weeks, or (iv) 12 mg every 4 weeks. Plasma uricase activity, plasma urate, and anti-pegloticase antibodies were measured, pharmacokinetic parameters were assessed, and adverse events were recorded. The mean plasma urate level was reduced to less than or equal to 6 mg/dl within 6 hours in all dosage groups, and this was sustained throughout the treatment period in the 8 mg and 12 mg dosage groups. The most effective dosage was 8 mg every 2 weeks. Twenty-six patients received all protocol

doses. The percentage of the patients in whom the primary efficacy end point (plasma urate less than 6 mg/dl for 80 % of the study period) was achieved ranged from 50 % to 88 %. Gout flares occurred in 88 % of the patients. The majority of adverse events (excluding gout flare) were unrelated to treatment and were mild or moderate in severity. Infusion-day adverse events were the most common reason for study withdrawal (12 of 15 withdrawals). There were no anaphylactic reactions. Anti-pegloticase antibody, present in 31 of 41 patients, was associated with reduced circulating half-life of pegloticase in some patients. The authors concluded that pegloticase, administered in multiple doses, was effective in rapidly reducing and maintaining plasma urate levels at less than or equal to 6 mg/dl in most patients in whom conventional therapy had been unsuccessful due to lack of response, intolerability, or contraindication.

Hershfield et al (2010) noted that a high plasma urate concentration (PUA), related to loss of urate oxidase in evolution, is postulated to protect humans from oxidative injury. This hypothesis has broad clinical relevance, but support rests largely on in vitro data and epidemiologic associations. Pegloticase therapy generates H_2O_2 while depleting urate, offering an in vivo test of the antioxidant hypothesis. These researchers showed that erythrocytes can efficiently eliminate H_2O_2 derived from urate oxidation to prevent cell injury in vitro; during therapy, disulfide-linked peroxiredoxin 2 dimer did not accumulate in red blood cells, indicating that their peroxidase capacity was not exceeded. To assess oxidative stress, these researchers monitored F2-isoprostanes (F2-isoPs) and protein carbonyls (PC), products of arachidonic acid and protein oxidation, in plasma of 26 refractory gout patients receiving up to 5 infusions of pegloticase at 3-week intervals. At baseline, PUA was markedly elevated in all patients, and plasma F2-isoP concentration was elevated in most. Pegloticase infusion rapidly lowered mean PUA to less than or equal to 1 mg/dL in all patients, and PUA remained low in 16 of 21 patients who completed treatment. F2-isoP levels did not correlate with PUA and did not increase during 15 weeks of sustained urate depletion. There also was no significant change in the levels of plasma PC. Because refractory gout is associated with high oxidative stress in spite of high PUA, and profoundly depleting uric acid did not increase lipid or protein oxidation, the authors concluded that urate is not a major factor controlling oxidative stress in vivo.

Several pipeline drugs for the treatment of gout include the selective uricosuric drug RDEA594 and various interleukin-1 (IL-1) inhibitors (anakinra, rilonacept, and canakinumab) (Burns and Wortmann, 2011). So et al (2007) stated that monosodium urate crystals stimulate monocytes and macrophages to release IL-1 beta via the NALP3 component of the inflammasome. The effectiveness of IL-1 inhibition in patients with hereditary auto-inflammatory syndromes with mutations in the NALP3 protein suggested that IL-1 inhibition might also be effective in relieving the inflammatory manifestations of acute gout. The effectiveness of IL-1 inhibition was first evaluated in a mouse model of monosodium urate crystal-induced inflammation. Inhibition of IL-1 prevented peritoneal neutrophil accumulation but tumor necrosis factor blockade had no effect. Based on these findings, these investigators performed a pilot, open-labeled study in 10 patients with gout who could not tolerate or had failed standard anti-inflammatory therapies. All patients received 100 mg anakinra daily for 3 days. All 10 patients with acute gout responded rapidly to anakinra. No adverse effects were observed. Blockade of IL-1 appears to be an effective therapy for acute gouty arthritis. The authors stated that these findings need to be confirmed in a controlled study.

In an observational study, Krishnan and colleagues (2012) examined if blood lead levels (BLLs) within the range currently considered acceptable are associated with gout. A total of 6,153 civilians aged 40 years or older with an estimated glomerular filtration rate greater than 10 ml/min per 1.73 m² were included in this study. Outcome variables were self-reported physician diagnosis of gout and serum urate level. Blood lead level was the principal exposure variable. Additional data collected were anthropometric measures, blood pressure, dietary purine intake, medication use, medical history, and serum creatinine concentration. The prevalence of gout was 6.05% (95 % confidence interval [CI]: 4.49 % to 7.62 %) among patients in the highest BLL quartile (mean of 0.19 μ mol/L [3.95 μ g/dL]) compared with 1.76 % (CI: 1.10 % to 2.42 %) among those in the lowest quartile (mean of 0.04 μ mol/L [0.89 μ g/dL]). Each doubling of BLL was associated with an unadjusted odds ratio of 1.74 (CI: 1.47 to 2.05) for gout and 1.25 (CI: 1.12 to 1.40) for hyperuricemia. After adjustment for renal function, diabetes, diuretic use, hypertension, race, body mass index, income, and education level, the highest BLL quartile was associated with a 3.6-fold higher risk for gout and a 1.9-fold higher

risk for hyperuricemia compared with the lowest quartile. The authors concluded that blood lead levels in the range currently considered acceptable are associated with increased prevalence of gout and hyperuricemia. The main drawback of this study was that blood lead level does not necessarily reflect the total body lead burden.

The updated European League Against Rheumatism (EULAR) guideline for the diagnosis and management of gout and hyperuricemia (Hamburger et al, 2011) did not mention testing for BLL. Furthermore, a University of Texas at Austin School of Nursing's clinical practice guideline on "Management of chronic gout in adults" (2012) do not mention measurement of BLL as a diagnostic tool.

In a Cochrane review, Sivera et al (2014) evaluated the benefits and harms of IL-1 inhibitors in acute gout. These investigators searched The Cochrane Library, MEDLINE and EMBASE on June 19, 2013. They applied no date or language restrictions. They performed a hand-search of the abstracts from the European League Against Rheumatism (EULAR) (2009 to 2012) and American College of Rheumatology (ACR) (2009 to 2011) conferences and of the references of all included trials. They also screened the Clinical Trials Registry Platform of the World Health Organization and Clinical Trials Registry Platform of the US National Institutes of Health. These researchers included randomized controlled trials (RCTs) and quasi-randomized clinical trials (controlled clinical trials (CCTs)) assessing an IL-1 inhibitor (e.g., anakinra, canakinumab or rilonacept) against placebo or another active treatment (colchicine, paracetamol, NSAIDs, glucocorticoids (systemic or intra-articular), adrenocorticotropin hormone, a different IL-1 blocking agent or a combination of any of the above) in adults with acute gout. Two review authors independently selected trials for inclusion, assessed the risk of bias and extracted the data. If appropriate, they pooled data in a meta-analysis. They assessed the quality of the evidence using the Grades of Recommendation, Assessment, Development, and Evaluation (GRADE) approach. These investigators included 4 studies (806 participants) in the review. The studies had an unclear risk of selection bias and low risk of performance and attrition biases. One study each had an unclear risk of detection and selection bias. Three studies (654 participants) compared subcutaneous canakinumab compared with intramuscular triamcinolone acetonide 40 mg in the treatment of acute gout flares of no

more than 5-day duration. Doses of canakinumab were varied (10 to 150 mg), but most people (255/368) were treated with canakinumab 150 mg. None of the studies provided data on participant-reported pain relief of 30 % or greater. Moderate-quality evidence indicated that canakinumab 150 mg was probably superior to triamcinolone acetonide 40 mg in terms of pain relief, resolution of joint swelling and in achieving a good treatment response at 72 hours following treatment, but was probably associated with an increased risk of adverse events (AEs). Mean pain (0- to 100-mm visual analog scale (VAS), where 0 mm was no pain) was 36 mm after triamcinolone acetonide treatment; pain was further reduced by a mean of 11 mm with canakinumab treatment (mean difference (MD) -10.6 mm, 95 % CI: -15.2 to -5.9). Forty-four per cent of participants treated with canakinumab had resolution of joint swelling at 72 hours compared with 32 % of participants treated with triamcinolone (risk ratio [RR] 1.39, 95 % CI: 1.11 to 1.74, number needed to treat for an additional beneficial outcome (NNTB) 9); 65 % of participants treated with canakinumab assessed their response to treatment as good or excellent compared with 47 % of participants treated with triamcinolone acetonide (RR 1.37, 95 % CI: 1.16 to 1.61, NNTB 6). Function or health-related quality of life (QOL) was not measured. In both groups, 0.7 % of participants withdrew from treatment (RR 1.1, 95 % CI: 0.2 to 7.2); there was 1 death and 1 alteration of laboratory results in each of the treatment groups. Adverse events were more frequent in participants receiving canakinumab (61%) compared with triamcinolone acetonide (51%; RR 1.2, 95% CI 1.1 to 1.4, number needed to treat for an additional harmful outcome (NNTH) 10). Low-quality evidence from one study (152 participants with an acute gout flare of no more than 48 hours' duration and affecting fewer than 4 joints) comparing rilonacept 320 mg with indomethacin (50 mg 3 times a day for 3 days followed by 25 mg 3 times a day for up to 9 days) indicated that indomethacin may improve pain more than rilonacept at 24 to 72 hours, and there may be no evidence of a difference in withdrawal rates or AEs. The mean change (improvement) in pain from baseline with indomethacin was 4.3 points (measured on a 0 to 10 numerical rating scale, where 0 was no pain); pain was improved by a mean of only 2.5 points with rilonacept (MD 2.52, 95 % CI: 0.29 to 4.75, 25 % less improvement in absolute pain with rilonacept). Inflammation, function health-related QOL and participant global assessment of treatment success were not measured. Rates of study withdrawals due to AEs were low in both groups: 1/75 (1 %)

participants in the rilonacept group compared with 2/76 (3 %) participants in the indomethacin group (RR 0.5, 95 % CI: 0.05 to 5.5). Adverse events were reported in 27/75 (36 %) participants in the rilonacept group and 23/76 (30 %) in the indomethacin group (RR 1.2, 95 % CI: 0.8 to 1.9). The authors concluded that moderate-quality evidence indicated that compared with a single suboptimal 40-mg dose of intramuscular injection of triamcinolone acetonide, a single subcutaneous dose of 150 mg of canakinumab probably results in better pain relief, joint swelling and participant-assessed global assessment of treatment response in people with an acute gout flare; but is probably associated with an increased risk of AEs. The cost of canakinumab is over 5,000 times higher than triamcinolone acetonide; however, there are no data on the cost-effectiveness of this approach. Moreover, the authors found no studies comparing canakinumab with more commonly used first-line therapies for acute gout flares such as NSAIDs or colchicine. Low-quality evidence indicated that compared with maximum doses of indomethacin (50 mg 3 times a day), 320 mg of rilonacept may provide less pain relief with a similar rate of AEs.

The Spanish Society of Rheumatology's clinical practice guidelines for "Management of gout" (SER, 2013) stated the following:

- It is not recommended to perform plain radiography, computed tomography (CT) or magnetic resonance imaging (MRI) for the diagnosis of gout (Level of evidence [LE] 2b; Grade of recommendation [GR] B).
- Ultrasound assists in the diagnosis of gout; crystal visualization is what establishes the definitive diagnosis (LE 4; GR C).
- Ultrasound-guided puncture facilitates obtaining fluid or other samples for the diagnosis of gout (LE 4; GR C).

An UpToDate review on "Clinical manifestations and diagnosis of gout" (Becker, 2014) states that "Ultrasound examination directed to joints or soft tissue deposits is an increasingly promising modality for the early detection and monitoring of therapy for gout".

Villaverde et al (2014) performed a systematic literature review of the usefulness of MRI and ultrasound (US) on assessment of treatment response in patients with gout. MEDLINE, EMBASE, Cochrane Library

(up to February 2012), and abstracts presented at the 2010 and 2011 meetings of the American College of Rheumatology and European League Against Rheumatism, were searched for treatment studies of any duration and therapeutic options, examining the ability of MRI/US to assess treatment response in gouty patients. Meta-analyses, systematic reviews, randomized clinical trials, cohort and case-control studies and validation studies were included. Quality was appraised using validated scales. There were only 3 US published studies in the literature that analyzed US utility on assessment of response to treatment in patients with gout. All of them were prospective case studies with a small number of patients and they were reviewed in detailed. A total of 36 patients with gout were examined with US. All of them had a baseline serum urate greater than 6 mg/dL. Ultrasound features of gout (double contour sign [DCS], hyper-echoic spots in synovial fluid, hyper-echoic cloudy areas, tophus diameter and volume) achieved significant reduction in patients who reached the objective of uricemia less than or equal to 6mg/dL in all the studies; however, patients in whom levels did not drop below 6 mg/dL had no change of US features of gout. Other parameters evaluated in 1 study included erythrocyte sedimentation rate (ESR), C-reactive protein (CRP), number of tender joints (TRN), number of swollen joints, and pain score (SP). All of them decreased with uricemia reduction, but only TRN and SP were statistically significant. No data were found on the value of MRI on treatment response assessment in patients with gout. The authors concluded that the improvement in US features showed concurrent validity with uric acid reduction. According to the published evidence, US can be a useful tool for monitoring treatment of gouty patients, although more research is needed. The value of MRI on treatment response assessment in patients with gout remains to be determined.

Ogdie et al (2015) examined the usefulness of imaging modalities in the classification of gout when compared to mono-sodium urate (MSU) crystal confirmation as the gold standard, in order to inform development of new gout classification criteria. These researchers systematically reviewed the published literature concerning the diagnostic performance of plain film radiography, MRI, US, conventional CT and dual energy CT (DECT). Only studies with MSU crystal confirmation as the gold standard were included. When more than 1 study examined the same imaging feature, the data were pooled and summary test characteristics were

calculated. A total of 11 studies (9 manuscripts and 2 meeting abstracts) satisfied the inclusion criteria. All were set in secondary care, with mean gout disease duration of at least 7 years. Three features were examined in more than 1 study: (i) DCS on US, (ii) tophus on US, and (iii) MSU crystal deposition on DECT. The pooled (95 % CI) sensitivity and specificity of US DCS were 0.83 (0.72 to 0.91) and 0.76 (0.68 to 0.83), respectively; of US tophus, were 0.65 (0.34 to 0.87) and 0.80 (0.38 to 0.96), respectively; and of DECT, were 0.87 (0.79 to 0.93) and 0.84 (0.75 to 0.90), respectively. The authors concluded that US and DECT show promise for gout classification; but the few studies to-date have mostly been in patients with longstanding, established disease. Moreover, they stated that the contribution of imaging over clinical features for gout classification criteria requires further examination.

An UpToDate review on "Treatment of acute gout" (Becker and Gaffo, 2019) state that the use of rilonacept for the "treatment" of gout flares remains investigational in the United States. In addition, a review in UpToDate on "Pharmacologic urate-lowering therapy and treatment of tophi in patients with gout" (Becker and Perez-Ruiz, 2019) state that "identification of IL-1 as a major cytokine in the initiation of gout flares has prompted interest in potential roles for IL-1 inhibitory agents (e.g., anakinra, canakinumab, or rilonacept) both in treatment of ongoing flares and as prophylaxis to prevent gout flares during the initiation of urate-lowering therapy. The role of these biologic agents in routine clinical practice remains to be defined". Canakinumab has been trialed as an "off-label use" for the treatment of refractory gout flares. Canakinumab was approved in the European Union for treatment of patients with at least three gout flares annually that cannot be effectively managed with other anti-inflammatory treatment options. Becker and Gaffo (2019) cite two identically designed randomized trials (one in the U.S., one in Europe) comparing canakinumab (single dose of 150 mg subcutaneous (SC)) plus intramuscular (IM) placebo to triamcinolone acetonide (40 mg IM) plus placebo administered SC (total n=456). Those trials found that canakinumab administration resulted in a significantly greater reduction in mean 72-hour pain score using a 100 mm visual analog scale. Four patients receiving canakinumab required hospitalization for treatment of infections; however, none were found to be opportunistic. Other adverse events that were most common with canakinumab included low

neutrophil counts and low platelet counts. Per Lexicomp, additional data may be necessary to further define the role of canakinumab in treatment of acute gout flares. Furthermore, Becker and Gaffo (2019) state that although canakinumab has shown efficacy in the treatment of gout flares, its use for this indication remains investigational in the United States.

The "2020 American College of Rheumatology Guideline for the Management of Gout" provides the following recommendations (not an all-inclusive list):

- For patients experiencing a gout flare, the panel strongly recommend using oral colchicine, NSAIDs, or glucocorticoids (oral, intraarticular, or intramuscular) as appropriate first-line therapy for gout flares over IL-1 inhibitors or ACTH (the choice of colchicine, NSAIDs, or glucocorticoids should be made based on patient factors and preferences). When colchicine is the chosen agent, the panel strongly recommend low-dose colchicine over high-dose colchicine given its similar efficacy and fewer adverse effects [high certainty of evidence];
- For patients experiencing a gout flare for whom other antiinflammatory therapies are poorly tolerated or contraindicated, the panel conditionally recommend using IL-1 inhibition over no therapy (beyond supportive/analgesic treatment) [moderate certainty of evidence];
- For patients who may receive NPO, the panel strongly recommend glucocorticoids (intramuscular, intravenous, or intraarticular) over IL-1 inhibitors or ACTH [high certainty of evidence];
- Note the recommendation with high quality of evidence includes network meta-analyses supporting canakinumab, which has superior mean pain score reduction and mean day-2 joint tenderness reduction. However, the Voting Panel raised concern that the comparator was weak (triamcinolone 40 mg) and that cost issues significantly favor other agents;
- For patients experiencing a gout flare, the panel conditionally recommend using topical ice as an adjuvant treatment over no adjuvant treatment [low certainty of evidence];
- The panel strongly recommend against pegloticase as first-line therapy. Moderate evidence is in support of the efficacy of pegloticase, but due to cost, safety concerns, and favorable

benefit- to- harm ratios of other untried treatment options, the recommendation is against using pegloticase as first- line agent;

- For patients with gout where xanthine oxidase inhibitor (XOI), uricosurics, and other interventions have failed to achieve serum urate (SU) target and who have frequent gout flares or nonresolving subcutaneous tophi, the panel strongly recommend switching to pegloticase over continuing current urate-lowering therapy (ULT). There is moderate certainty of evidence about the efficacy of the benefits, harms, and high certainty about the costs of pegloticase. For patients with high disease activity, the magnitude of potential benefits outweighs the harms and costs of the drug;
- For patients with gout for whom XOI, uricosurics, and other interventions have failed to achieve SU target and who have infrequent gout flares (less than 2 flares/year) and no tophi, the panel strongly recommend against switching to pegloticase over continuing current ULT [moderate certainty of evidence] For patients with minimal disease activity, the smaller potential benefits do not outweigh the harms and costs of the drug.

Alpha-1-Antitrypsin-Fc Fusion Protein for the Treatment of Gout

Joosten et al (2016) generated a new protein, recombinant human alpha-1-antitrypsin (AAT)-IgG1 Fc fusion protein (AAT-Fc), and evaluated its properties to suppress inflammation and IL-1 β in a mouse model of gouty arthritis. A combination of MSU crystals and the fatty acid C16.0 (MSU/C16.0) was injected intra-articularly into the knee to induce gouty arthritis. Joint swelling, synovial cytokine production and histopathology were determined after 4 hours; AAT-Fc was evaluated for inhibition of MSU/C16.0-induced IL-1 β release from human blood monocytes and for inhibition of extracellular IL-1 β precursor processing. AAT-Fc markedly suppressed MSU/C16.0-induced joint inflammation by 85 to 91 % ($p < 0.001$). Ex-vivo productions of IL-1 β and IL-6 from cultured synovia were similarly reduced (63 % and 65 %, respectively). The efficacy of 2.0 mg/kg AAT-Fc in reducing inflammation was comparable to 80 mg/kg of plasma-derived AAT. Injection of AAT-Fc into mice increased circulating levels of endogenous IL-1 receptor antagonist by 4-fold. These investigators also observed that joint swelling was reduced by 80 %, cellular infiltration by 95 % and synovial production of IL-1 β by 60 % in

transgenic mice expressing low levels of human AAT. In-vitro, AAT-Fc reduced MSU/C16.0-induced release of IL-1 β from human blood monocytes and inhibited proteinase-3-mediated extracellular processing of the IL-1 β precursor into active IL-1 β . The authors concluded that a single low dose of AAT-Fc is highly effective in reducing joint inflammation in this murine model of acute gouty arthritis. They stated that considering the long-term safety of plasma-derived AAT use in humans, subcutaneous AAT-Fc emerges as a promising therapy for gout attacks.

Baseline Gut Microbiome as a Predictive Biomarker of Response to Probiotic Adjuvant Treatment in the Management of Gout

Zhao et al (2024) noted that gout is characterized by dysregulation of UA metabolism, and the gut microbiota may serve as a regulatory target. In a 2-month randomized, double-blind, placebo-controlled, multi-center study, these investigators examined the additional benefits of co-administration of Probio-X alongside febuxostat. A total of 160 patients with gout were randomly assigned to either the probiotic group (n = 120; Probio-X [1×10^{11} CFU/day] with febuxostat) or the placebo group (n = 40; placebo material with febuxostat). Co-administration of Probio-X significantly decreased serum UA levels and the rate of acute gout attacks ($p < 0.05$). Based on achieving a target sUA level (360 $\mu\text{mol/L}$) after the intervention, the probiotic group was further sub-divided into probiotic-responsive (ProA; n = 54) and probiotic-unresponsive (ProB; n = 66) sub-groups. Post-intervention clinical indicators, metagenomic, and metabolomic changes in the ProB and placebo groups were similar, but differed from those in the ProA group, which exhibited significantly lower levels of acute gout attack, gout impact score, serum indicators (UA, xanthine oxidase [XOD], hypoxanthine, and IL-1 β), and fecal gene abundances of UA-producing pathways (KEGG orthologs of K13479 and K01487; gut metabolic modules for formate conversion and lactose and galactose degradation). Furthermore, the ProA group showed significantly higher levels ($p < 0.05$) of gut SCFAs-producing bacteria and UA-related metabolites (xanthine, hypoxanthine, bile acids) following the intervention. Lastly, these investigators established a gout metagenomic classifier to predict probiotic responsiveness based on subjects' baseline gut microbiota composition. The authors concluded that the higher clinical response rate in the probiotic group compared to the placebo group suggested the effectiveness of Probio-X in the treatment of gout.

Moreover, these researchers stated that the individualized responses towards probiotic intervention could be related to the subjects' endogenous gut microbiota composition; this study provided valuable information for future studies aiming to examine the use of probiotic in the management of patients with gout.

The authors stated that this trial had several drawbacks. First, the study population consisted solely of Chinese patients, limiting the generalizability of the results to patients with gout from other countries, ethnicities, and clinical backgrounds. Second, due to patient preference and cost considerations, these researchers did not evaluate the CT results of the gout site. Third, these investigators did not carry out follow-up assessments after the intervention to examine the long-term effects of Probio-X on lowering sUA levels.

Digital Tomosynthesis

Son and colleagues (2017) compared 3 radiographic methods: (i) digital tomosynthesis (DT), (ii) plain radiography, and (iii) computed tomography (CT) for evaluating changes in feet of patients with chronic gouty arthritis. Two independent radiologists read the plain radiography, DT, and CT images of 30 male patients with gout. The degrees of erosion and joint space narrowing were scored using the Sharp-van der Heijde scoring method in 18 foot joints, which consisted of 4 PIP and 1 IP joint of the 1st toe, 5 MP, 5 tarsometatarsal, and 3 naviculo-cuneiform joints of the foot. DT showed high reproducibility [0.929 for intra-observer intra-class correlation coefficient (ICC) and 0.838 for inter-observer ICC]. DT showed similar results to those of CT and superior results to those of plain radiography for evaluating radiographic damage [mean total score, 8.5 ± 14.6 ($\pm$ standard deviation) for plain radiography, 12.9 ± 12.4 for DT, and 12.6 ± 11.2 for CT]. The authors concluded that the findings of this study showed that DT is a good method for evaluating radiographic changes in patients with gout. Moreover, they stated that further research is needed to apply DT to actual clinical settings.

Dual-Energy Computed Tomography-Based Radiomics for Differentiating Patients With and Without Gout Flares

Hu et al (2024) stated that currently there is a lack of data regarding the potential link between gout flares and DECT radiomic features. These researchers constructed and validated a comprehensive DECT-based radiomics model for differentiating patients with and without gout flares.

The analysis included 200 patients, of whom 150 were confirmed to have experienced at least 1 flare in the past 12 months; the remaining 50 patients did not experience flares. The radiomic features of the tophi at the bilateral 1st metatarsophalangeal joints were extracted and analyzed.

Optimal radiomic features were selected using the least absolute shrinkage and selection operator method, and logistic regression analysis was employed to screen clinical characteristics and establish a clinical model. The optimal radiomic features were then combined with the identified independent clinical variables to develop a comprehensive model. The performances of the radiomic, clinical, and comprehensive models were examined using ROC analysis, calibration curves, and decision curve analysis. A total of 4 radiomic features distinguished patients with at least 1 flare from those without flares and were used to establish the radiomic model. Disease duration and hypertension were independent factors that differentiated flare occurrences. The radiomic, clinical, and comprehensive models showed favorable discrimination, with areas under the receiver operating characteristic curves (AUROC) of 0.76 (95 % CI: 0.69 to 0.83), 0.72 (95 % CI: 0.63 to 0.80), and 0.79 (95 % CI: 0.73 to 0.86), respectively. The calibration curves ($p > 0.05$) showed that the differentiated values of the comprehensive model agreed well with the actual values. Decision curve analysis showed that the comprehensive model achieved higher net clinical benefits than the use of either the radiomic or clinical model alone. The authors concluded that the findings of this study suggested that a radiomics model could distinguish patients with and without gout flares. The proposed clinical radiomics nomogram could increase the effectiveness of differentiating flare occurrence, which may facilitate the clinical decision-making process. These findings need to be validated by well-designed studies.

Dual-Energy Computed Tomography for the Diagnosis of Recent Onset Gout

Yu and colleagues (2018) noted that many clinicians tend to use dual-energy computed tomography (DECT) instead of aspiration biopsy in the diagnosis of gout, but its accuracy has shown controversial results. In

this systematic review and meta-analysis, these researchers examined the accuracy of DECT in the diagnosis of gout. They performed a systematic review of the literature published in Medline, Embase, PubMed, and Cochrane databases. Studies included are all clinical trials of DECT in the diagnosis of gout. Quality assessment of bias and applicability was conducted using the Quality of Diagnostic Accuracy Studies-2 (QUADAS-2). These investigators recorded sensitivity and specificity of algorithms and calculated PLR, NLR and DOR, and respective CIs. The summary ROC (sROC) was drawn to get the Cochran Q-index and the AUC. A total of 7 studies were included in this review and showed high homogeneity. The analysis results presented the pooled sensitivity was 88 % (95 % CI: 84 to 90 %) and specificity was 90 % (95 % CI: 85 to 93 %). Then, these researchers figured out that the pooled PLR was 8.48 (95 % CI: 5.89 to 12.22) and NLR was 0.10 (95 % CI: 0.04 to 0.24) respectively. In addition, Cochran-Q was 0.90 and AUC was 0.9565 in sROC curve. The authors concluded that DECT showed relatively high sensitivity and specificity in the diagnosis of gout. Synthetically considering these DECT abnormalities could improve the diagnostic sensitivity. Moreover, these researchers stated that more rigorous and standardized studies are still needed to support these findings.

In a systematic review and meta-analysis, Gamala and associates (2019) examined the utility of DECT for diagnosing gout. These researchers carried out a systematic literature search in PubMed, Embase and Cochrane Library. Studies evaluating the utility of DECT for diagnosing gout were included. Reference standards were detection of monosodium urate crystals at SF assessment or a validated set of criteria. The methodological quality of studies was evaluated according to the QUADAS-2 criteria. Data from person-based and joint-/localization-based evaluations were pooled separately, and subgroup analyses for disease stage/duration and reference standard were performed. A total of 10 studies were included; in person-based evaluations, the pooled (95 % CI) sensitivity and specificity were 0.81 (0.77 to 0.86) and 0.91 (0.85 to 0.95), respectively. In joint-based evaluations, they were 0.83 (0.79 to 0.86) and 0.88 (0.83 to 0.92), respectively. At short disease duration (less than or equal to 6 weeks), the pooled (95 % CI) sensitivity and specificity at the joint level were 0.55 (0.46 to 0.64) and 0.89 (0.84 to 0.94), respectively. The authors concluded that DECT had a high

diagnostic accuracy in established gout, but its diagnostic sensitivity was low in subjects with recent onset gout.

Furthermore, an UpToDate review on "Clinical manifestations and diagnosis of gout" (Gaffo, 2019) states that "We limit the use of DECT examination for gout diagnosis to patients in whom, despite more chronic arthropathy or deformity, a urate crystal deposition basis for the causative disorder has not been confirmed by polarized light microscopic examination of joint aspirates, pathologic analysis of tissue samples, or alternative imaging modalities, including magnetic resonance imaging (MRI)".

In a systematic review, Stauder and Peloso (2022) examined if there is clinical utility for DECT to inform on prognosis for patients with gout. With DECT, individualized treatment plans could be developed based on the patient's unique urate burden, with DECT being used as a clinical outcome measure in gout management. To evaluate DECT as a reliable, valid, and sensitive prognostic instrument, a librarian-assisted search was carried out in PubMed and Embase for studies on gout and DECT informing on reliability; content, construct, and criterion validity; sensitivity to change; and minimum clinically important changes. This systematic literature review showed that DECT has high intra- and inter-rater reliability. Tophus burden correlated with functional loss to show content validity. DECT volume was positively correlated with death, cardiovascular risk factors, and the risk for future gout flares. DECT exhibited excellent sensitivity to change with effective urate-lowering therapies. The authors concluded that DECT is a promising prognostic tool based on its high reliability, sensitivity to change, and emerging validity. Moreover, these researchers stated that additional prospective, properly powered, well-designed studies are needed to examine its prognostic utility. This systematic review suggested that DECT very likely has additional prognostic information beyond clinical tophi assessment alone.

The authors stated that this systematic review had several drawbacks. First, the conclusions were limited by the quantity and quality of the current literature. Most of the literature relating to DECT reliability has shown it to be excellent. The diagnostic performance of DECT and US in early gout requires more study where DECT sensitivity appears lower.

Second, the prognostic value of DECT abnormalities in the absence of clinical tophi is not fully defined but is important, as up to 50.0 % of patients with gout have an abnormal DECT scan without clinical tophi or abnormal serum urate levels. Third, these investigators found limited literature on the relationship of DECT to disability; and they also found limited literature on the minimum important change in DECT.

Genetic Testing

Matsuo and colleagues (2016) stated that although genome-wide association studies (GWASs) of gout have been reported, they included self-reported gout cases in which clinical information was insufficient. Thus, the relationship between genetic variation and clinical subtypes of gout remains unclear. These researchers performed a GWAS of clinically defined gout cases only. A GWAS was conducted with 945 patients with clinically defined gout and 1,213 controls in a Japanese male population, followed by replication study of 1,048 clinically defined cases and 1,334 controls. Five gout susceptibility loci were identified at the genome-wide significance level ($p < 5.0 \times 10^{-8}$), which contained well-known urate transporter genes (ABCG2 and SLC2A9) and additional genes: rs1260326 ($p = 1.9 \times 10^{-12}$; odds ratio [OR] = 1.36) of GCKR (a gene for glucose and lipid metabolism), rs2188380 ($p = 1.6 \times 10^{-23}$; OR = 1.75) of MYL2-CUX2 (genes associated with cholesterol and diabetes mellitus) and rs4073582 ($p = 6.4 \times 10^{-9}$; OR = 1.66) of CNIH-2 (a gene for regulation of glutamate signaling). The latter 2 were identified as novel gout loci. Furthermore, among the identified single-nucleotide polymorphisms (SNPs), these investigators demonstrated that the SNPs of ABCG2 and SLC2A9 were differentially associated with types of gout and clinical parameters underlying specific subtypes (renal under-excretion type and renal over-load type). The effect of the risk allele of each SNP on clinical parameters showed significant linear relationships with the ratio of the case-control ORs for 2 distinct types of gout ($r = 0.96$ [$p = 4.8 \times 10^{-4}$] for urate clearance and $r = 0.96$ [$p = 5.0 \times 10^{-4}$] for urinary urate excretion). The authors concluded that they conducted the first GWAS using patients with clinically defined gout only and identified 5 loci containing 2 novel loci. Moreover, identified SNPs showed differential effects on different gout types and affected clinical parameters underlying specific types. Thus, genetic testing for gout may well be introduced into future companion diagnostics. For example, patients with

risk alleles for renal over-load (ROL)-type gout would be given urate synthesis inhibitors such as allopurinol and febuxostat, while patients with risk alleles for renal under-excretion (RUE)-type gout would be administered uricosuric agents including benzbromarone and lesinurad, a selective uric acid reabsorption inhibitor that has just finished its phase III study. They stated that exploring genetic heterogeneity among different gout types will deepen understanding of the etiology of gout and serve to categorize patients for future personalized treatment.

Dalbeth and associates (2017) noted that over the past 10 years, there have been major advances in the understanding of the genetic basis of hyperuricemia and gout as well as of the pharmacogenetics of urate-lowering therapy (ULT). Key findings included the reporting of 28 urate-associated loci, the discovery that ABCG2 plays a central role on extra-renal uric acid excretion, the identification of genes associated with development of gout in the context of hyperuricemia, recognition that ABCG2 variants influence allopurinol response, and the impact of HLA-B*5801 testing in reducing the prevalence of allopurinol hypersensitivity in high-risk populations. These advances, together with the reducing cost of whole genome sequencing, mean that integrated personalized medicine approaches may soon be possible in clinical practice. Genetic data may inform assessment of disease prognosis in individuals with hyperuricemia or established gout, personalized lifestyle advice, selection and dosing of ULT, and prevention of serious AEs. The authors concluded that rapidly progressive technology and disease-specific genetic discoveries have the potential to make personalized medicine a reality in many aspects of gout management, including risk assessment of disease progression, personalized lifestyle advice, selection and dosing of ULT, and prevention of serious AEs. They stated that although major progress has been made through GWAS, there is a further need for large, well-characterized datasets that include different disease states, detailed pharmacology (including dose information, therapeutic response, AEs) and lifestyle information. They noted that a further challenge is population-specific effects, meaning that discoveries in one population may not be translatable to other populations. In order to avoid increasing the disparities that are already evident in gout management, study of different populations will be essential, particularly of those with high prevalence of severe disease.

Cleophas and co-workers (2017) stated that as a result of the association of a common polymorphism (rs2231142, Q141K) in the ATP-binding cassette G2 (ABCG2) transporter with serum urate concentration in a GWAS, it was revealed that ABCG2 is an important uric acid transporter.

These investigators discussed the relevance of ABCG2 polymorphisms in gout, possible etiological mechanisms, and therapeutic approaches.

The 141K ABCG2 urate-increasing variant causes instability in the nucleotide-binding domain, leading to decreased surface expression and function. Trafficking of the protein to the cell membrane is altered, and instead, there is an increased ubiquitin-mediated proteasomal degradation of the variant protein as well as sequestration into aggresomes. In humans, this resulted in decreased uric acid excretion through both the kidney and the gut with the potential for a subsequent compensatory increase in renal urinary excretion. Not only does the 141K polymorphism in ABCG2 lead to hyperuricemia through renal overload and renal under-excretion, but emerging evidence indicates that it also increases the risk of acute gout in the presence of hyperuricemia, early onset of gout, tophi formation, and a poor response to allopurinol.

In addition, there is some evidence that ABCG2 dysfunction may promote renal dysfunction in chronic kidney disease patients, increase systemic inflammatory responses, and decrease cellular autophagic responses to stress. The authors concluded that these findings suggested multiple benefits in restoring ABCG2 function. It has been shown that decreased ABCG2 141K surface expression and function can be restored with colchicine and other small molecule correctors.

However, caution should be exercised in any application of these approaches given the role of surface ABCG2 in drug resistance. These researchers noted that the ABCG2 transporter is an important molecule in urate excretion. Decreased ABCG2 expression and function due to genetic polymorphisms leads to both ROL hyperuricemia and RUE hyperuricemia. The most extensively studied genetic variant is Q141K.

Besides significantly increasing serum urate concentration, the 141K ABCG2 variant has also been associated with acute gout, tophaceous gout, and poor allopurinol response. In addition, 141K-induced hyperuricemia may lead to excessive inflammatory responses and decreased ABCG2 function may cause defective autophagy. They stated that all of these effects warrant further research to the restoration of 141K ABCG2 function and surface expression, for example, by small molecules.

Gut Microbiota Dysbiosis in the Management of Gout

Tong et al (2022) stated that many studies have reported that gut microbiota is important for the host's health because it regulates the host's metabolism, endocrine, and immune systems. In recent years, increasing evidence has demonstrated that gut microbiota plays a role in the onset and progression of gout. Changes in the composition and metabolism of the gut microbiota, result in abnormalities of UA degradation, increasing UA generation, releasing pro-inflammatory mediators, and intestinal barrier damage in developing gout. As a result, therapy that targets gut microbiota has drawn significant interest. These investigators examined how the gut microbiota contributes to the pathophysiology of gout and how gout affects the gut microbiota. Furthermore, they explained how gut microbiota might serve as a unique index for the diagnosis of gout and how conventional gout treatment medicines interact with it. The authors concluded that prospective, therapeutic approaches focusing on gut microbiota for the prevention and treatment of gout were highlighted, which may represent a future avenue in gout treatment.

Shirvani-Rad et al (2023) noted that gut dysbiosis is believed to be one of the several mechanisms that are involved in the pathogenesis of gout. In a systematic review, these investigators examined the role of gut dysbiosis in gout disease and uncover the underlying mechanisms. They carried out a comprehensive search on PubMed, Web of Science, and Scopus databases up to October 2021. Animal studies and human observational studies, including case-control, cross-sectional, and cohort studies examining the association between gut microbiota composition and gout were included. The quality of included studies was assessed using the Newcastle-Ottawa Quality Assessment scale (NOS) and the SYRCLE's risk of bias tool. These researchers found 274 studies among which 15 studies were included in this systematic review. Of them, 10 studies were carried out on humans and 5 studies were conducted on animals. Increased abundance of *Altiples* and decreased abundance of *Enterobacteriaceae* altered purine metabolism; thus, aggravating gout condition. Moreover, a higher abundance of *Phascolarctobacterium* and *Bacteroides* in gout modulates enzymatic activity in purine metabolism. Butyrate-producing bacteria such as *Faecalibacterium*, *Prausnitzii*, *Oscillibacter*, *Butyricoccus*, and *Bifidobacterium* had higher abundance in

healthy controls compared to gout patients, suggesting the anti-inflammatory and anti-microbial role of short-chain fatty acids (SCFAs). Lipopolysaccharides (LPS)-releasing bacteria, such as enterobacteriaceae, prevotella, and bacteroides, are also involved in the pathogenesis of gout disease by stimulating the innate immune system. The authors concluded that microbiota is deeply associated with inflammatory disorders like gout via many mechanisms. Some contradictory results prevent clinicians in determining the role of some special bacterial taxa. There was no clear evidence regarding which bacteria are more protective. Different designs and methodologies of studies could be a reason for those controversies. In general, studies were different in the eligibility criteria, sample size, diagnostic tools, microbiota and biochemical analysis techniques, which may warrant the controversies. These researchers stated that further studies are needed to improve gout through microbiota-modulating-based therapies.

The authors stated that diverse methods used for microbiota analysis, diagnostic tools, and outcome measures in different studies were the key drawbacks of this systematic review, which made it challenging to compare and combine the results. Furthermore, all included studies in this review were observational in nature; therefore, it was hard to establish assured causality but instead correlations were acknowledged. These researchers stated that further studies are needed to generate basic knowledge for clarifying possible underlying mechanisms and any probable causal relationship.

Indolelactic Acid as a Metabolic Biomarker for the Diagnosing of Gout

Zhang et al (2024) noted that gout is a heterogeneous disease caused by the deposition of monosodium urate crystals in joints; however, its pathogenesis is poorly understood. The discovery of novel biomarkers is necessary for the early detection and diagnosis of gout. These investigators characterized the metabolic profile of patients with gout using metabolomics, and uncovered the underlying pathological mechanisms resulting in gout development. Serum samples were collected from 49 healthy participants and 47 patients with gout. Using ultra-high-performance liquid chromatography (HPLC) Orbitrap Exploris mass spectrometer non-target metabolomics technology, with a variable

importance in the projection of greater than 1 and a false discovery rate adjusted $p < 0.05$ was used, while a biomarker panel was screened using ROC analysis. The potential differentially expressed markers related to gout were identified by ROC analysis, and the ESR, UA, alanine transaminase, aspartate aminotransferase, creatinine, triglyceride, total cholesterol, HDL and LDL levels were significantly different in the group of patients with gout compared with those in healthy individuals. A total of 186 differentially expressed metabolites were identified, with 156 differential metabolites up-regulated and 30 down-regulated in the patients with gout compared with healthy individuals. Pathway analysis revealed that D-glutamine and D-glutamate metabolism may serve key roles in gout. The authors concluded that compared with healthy individuals, the indolelactic acid (ILA) level of patients with gout was significantly higher. These investigators stated that ILA may serve as a potential biomarker for the diagnosis of gout and could be used to detect or predict gout progression in the future.

The authors stated that this study had several drawbacks. First, due to the demographics of the patients with gout, a large number of the patients included in this trial were younger than the healthy subjects and the majority of patients with gout were men. Thus, the 2 groups were not comparable in age or sex. Second, the small sample size ($n = 47$ gout patients) of the present study. The drawbacks of the findings of this trial were with regard to age, which was increased due to selection bias. The lack of data for patient height, weight, and BMI were also a drawback. As a consequence of the 2 groups not being randomized or matched, there was a high risk of selection bias. The ratio of 2:1 or 7:3 was employed to divide individuals into groups in previous studies; however, when processing the data in the early stage of this trial, it was found that the samples after 2:1 distribution were scattered; therefore, the ratio of 7:3 was selected to randomly distribute the samples. In the future, longitudinal studies should be conducted in combination with other omics studies, including proteomics and genomics, or machine learning (ML) to identify additional potential biomarkers and pathways involved in gout.

Lesinurad plus Allopurinol or Febuxostat for the Treatment of Tophi in Gout

Sriranganathan and colleagues (2021) noted that tophi develop in

untreated or uncontrolled gout. This is an update of a Cochrane Review first published in 2014. These researchers examined the benefits and harms of non-surgical and surgical treatments for the management of tophi in gout. They updated the search of Cochrane Central Register of Controlled Trials (CENTRAL), Medline and Embase databases to August 28, 2020. These investigators included all published RCTs or controlled clinical trials examining interventions for tophi in gout in adults. They used standard methodological procedures expected by Cochrane. These researchers included 1 trial in their original review. They added 4 more trials (1,796 subjects) in this update. One had 3 arms: pegloticase infusion bi-weekly, monthly pegloticase infusion (pegloticase infusion alternating with placebo infusion every 2 weeks) and placebo. Two studies examined lesinurad 200-mg or 400-mg in combination with allopurinol. One trial studied lesinurad 200-mg or 400-mg in combination with febuxostat. One trial compared febuxostat 80-mg and 120-mg to allopurinol. Two trials were at unclear risk of performance and detection bias due to lack of information on blinding of subjects and personnel. All other trials were at low risk of bias. Moderate-certainty evidence (downgraded for imprecision; 1 study; 79 subjects) showed that bi-weekly pegloticase resolved tophi in 21/52 subjects compared with 2/27 on placebo (RR 5.45, 95 % CI: 1.38 to 21.54; number needed to treat for a benefit (NNTB) 3, 95 % CI: 2 to 6). Similar proportions of subjects receiving bi-weekly pegloticase (80/85) had an AE compared to placebo (41/43) (RR 0.99, 95 % CI: 0.91 to 1.07). However, more subjects on bi-weekly pegloticase (15/85) withdrew due to an AE compared to placebo (1/43) (RR 7.59, 95 % CI: 1.04 to 55.55; number needed to treat for a harm (NNTH) 7, 95 % CI: 4 to 16). More subjects on monthly pegloticase (11/52) showed complete resolution of tophi compared with placebo (2/27) (RR 2.86, 95 % CI: 0.68 to 11.97; NNTB 8, 95 % CI: 4 to 91). Similar numbers of subjects on monthly pegloticase (84/84) had an AE compared to placebo (41/43) (RR 1.05, 95 % CI: 0.98 to 1.14). More subjects on monthly pegloticase (16/84) withdrew due to AEs compared to placebo (1/43) (RR 8.19, 95 % CI: 1.12 to 59.71; NNTH 6, 95 % CI: 4 to 14). Infusion reaction was the most common reason for withdrawal. Moderate-certainty evidence (2 studies; 103 subjects; downgraded for imprecision) showed no clinically significant difference for complete resolution of target tophus in the lesinurad 200-mg plus allopurinol arm (11/53) compared to the placebo plus allopurinol arm (16/50) (RR 0.40, 95 % CI: 0.04 to 4.57), or in the lesinurad 400-mg plus allopurinol arm

(12/48) compared to the placebo plus allopurinol arm (16/50) (RR 0.79, 95 % CI: 0.42 to 1.49). An extension study examined lesinurad 200-mg or 400-mg in combination with febuxostat, or placebo (low-certainty evidence, down-graded for indirectness and imprecision). Subjects on lesinurad in the original study continued (CONT) on the same dose. Lesinurad 400-mg plus febuxostat may be beneficial for tophi resolution; 43/65 in the lesinurad 400-mg CONT arm compared to 38/64 in the lesinurad 200-mg CONT arm had tophi resolution (RR 1.11, 95 % CI: 0.85 to 1.46). Lesinurad 400-mg plus febuxostat may result in no difference in AEs; 57/65 in the lesinurad 400-mg CONT arm had an AE compared to 50/64 in lesinurad 200-mg CONT arm (RR 1.12, 95 % CI: 0.96 to 1.32). Lesinurad 400-mg plus febuxostat may result in no difference in withdrawals due to AEs; 10/65 subjects in the lesinurad 400-mg CONT arm withdrew due to an AE compared to 10/64 subjects in the lesinurad 200-mg CONT arm (RR 0.98, 95 % CI: 0.44 to 2.20). Lesinurad 400-mg plus febuxostat may result in no difference in mean serum uric acid (sUA), which was 3 mg/dL in the lesinurad 400-mg CONT group compared to 3.9 mg/dL in the lesinurad 200-mg CONT group (MD -0.90, 95 % CI: -1.51 to -0.29). Subjects who were not on lesinurad in the original study were randomized (CROSS) to lesinurad 200-mg or 400-mg, both in combination with febuxostat. Low-certainty evidence down-graded for indirectness and imprecision showed that lesinurad 400-mg (CROSS) may result in tophi resolution (17/34) compared to lesinurad 200-mg (CROSS) (14/33) (RR 1.18, 95 % CI: 0.70 to 1.98). Lesinurad 400-mg in combination with febuxostat may result in no difference in AEs (33/34 in the lesinurad 400-mg CROSS arm compared to 27/33 in the lesinurad 200-mg (CROSS); RR 1.19, 95 % CI: 1.00 to 1.41). Lesinurad 400-mg plus febuxostat may result in no difference in withdrawals due to AEs, 5/34 in the lesinurad 400-mg CROSS arm withdrew compared to 2/33 in the lesinurad 200-mg CROSS arm (RR 2.43, 95 % CI: 0.51 to 11.64). Lesinurad 400-mg plus febuxostat results in no difference in sUA (4.2 mg/dL in lesinurad 400-mg CROSS) compared to lesinurad 200-mg (3.8 mg/dL in lesinurad 200-mg CROSS; MD 0.40 mg/dL, 95 % CI: -0.75 to 1.55). The authors concluded that moderate-certainty evidence showed that pegloticase was probably beneficial for resolution of tophi in gout. Although there was little difference in AEs when compared to placebo, subjects on pegloticase had more withdrawals due to AEs. Lesinurad 400-mg plus febuxostat may be beneficial for tophi resolution compared with lesinurad 200-mg plus febuxostat; there was no difference

in AEs between these groups. These researchers were unable to determine whether lesinurad plus febuxostat was more effective than placebo. Lesinurad (400-mg or 200-mg) plus allopurinol was probably not beneficial for tophi resolution, and there was no difference in AEs between these groups. These researchers stated that RCTs on interventions for managing tophi in gout are needed, and the lack of trial data is surprising given that allopurinol is a well-established treatment for gout.

Measurement of Serum Cystatin C Level as a Marker of the Renal Function Damage and Inflammation

Zhang and colleagues (2019) examined the changes of serum uric acid (sUA), lipids and cystatin C (CysC) in primary gout patients, and explored the clinical significance in gout patients. sUA, CysC, high-sensitivity CRP (hsCRP) and other biochemical parameters were measured in 326 gout patient and 210 healthy control subjects, blood cell counts were also detected. Clinical data were collected from gout patients. sUA, CysC, hsCRP, body mass index (BMI), white blood cell (WBC) counts, neutrophil granulocyte (GR), monocyte (Mo), triglycerides (TG), plasma total cholesterol (TC), very low density lipoprotein (VLDL), apolipoprotein B100 (apoB100), blood glucose (GLU), serum creatinine (sCr) and urea nitrogen (BUN) were significantly increased in gout patients compared with HC subjects ($p < 0.01$, respectively), while lymphocyte counts and high density lipoprotein-cholesterol (HDL-C) were significantly decreased in gout patients compared with HC subjects ($p < 0.01$, respectively).

Positive correlations were observed between concentration of sUA and age, TG, VLDL, sCr and CysC ($p < 0.05$, respectively); while negative correlations were observed between the concentration of sUA and HDL-C ($p < 0.01$). Besides, positive correlations were observed between concentration of CysC and WBC, GR, Mo, apoA1, GLU, sCr, BUN, sUA, hsCRP ($p < 0.05$, respectively); while negative correlations were observed between the concentration of CysC and TC, LDL-C ($p < 0.01$, respectively). The authors concluded that blood lipid profile was changed in gout patients. Gout patients who suffered from lipid metabolism disorder and vascular diseases might be associated with hyperuricemia, which led to endothelial cell damage and vascular smooth muscle cell proliferation. They stated that serum CysC level might be as a marker of the renal function damage and inflammation; hyperuricemia was the risk

factor of renal disorder in gout patients.

Measurement of Synovial Fluid Uric Acid Level for Diagnosis of Gout

Vaidya and colleagues (2018) stated that examination of urate crystal in synovial fluid (SF) remains the gold standard for diagnosis of gout, but is not universally available. Synovial fluid uric acid (UA) level may be measured by the uricase method with an automated analyzer. The present study aimed to evaluate the utility of SF to serum UA ratio (SSR) for diagnosis of gout. A cross-sectional study was conducted at the National Center for Rheumatic Diseases, Nepal. Patients presenting with acute (less than 1 day) joint pain and/or swelling were included.

Aspiration was performed in all patients and fluid was subjected to testing for urate level, pH and cell counts and microscopy. Serum samples were also assessed for urate levels, and the SSR was calculated for each patient. A receiver operating characteristic curve (ROC) was plotted to determine the cut-off value for indicating diagnosis of gout. The difference in SSR between gout and non-gout effusion was evaluated by 1-way analysis of variance. A total of 181 patients were included of which 77 had gout. The remaining cases included osteoarthritis, pseudo-gout, rheumatoid arthritis and ankylosing spondylitis; SSR was significantly higher in gout patients than in any other group ($p < 0.05$). An SSR of greater than or equal to 1.01 had the highest sensitivity and specificity at 89.6 % and 66.3 %, respectively, for identifying gout effusion. The authors concluded that these findings indicated that SSR may be used as an aid for gout diagnosis when polarizing microscopy is not available.

Metabolites as Potential Biomarkers for the Early Detection of Gout

Li et al (2023) noted that many studies have attempted to examine potential biomarkers for the early detection of gout; however, consistent and high levels of evidence are lacking. In a systematic review and meta-analysis, these investigators examined the potential value of metabolites in predicting the occurrence and development of gout. They searched the databases including the Embase, the Cochrane Library, PubMed, Web of Science, VIP Date, Wanfang Data, and CNKI; and the

screening was fulfilled on July 30, 2022. The records were screened according to the inclusion criteria and the risk of bias was assessed. Qualitative analysis was carried out for all metabolites, and meta-analysis was conducted for metabolite concentrations using random effects to calculate the standard mean difference (SMD) and 95 % CI. A total of 2,738 records were identified, 33 studies with 3,422 subjects were included, and 701 metabolites were identified. The qualitative analysis results showed that compared with the healthy control group, the concentration of 56 metabolites increased, and 22 metabolites decreased. The results of the meta-analysis indicated that 17 metabolites were statistically significant. The authors concluded that changes in metabolites in humans do shed light on the underlying mechanisms of gout development. Some specific metabolites such as UA, hypoxanthine, xanthine, kynurenic acid (KYNA), guanosine, adenosine, creatinine, leukotriene B4 (LB4), DL-2-aminoadipic acid have been highlighted in the development of gout, establishing the potential of metabolites as predictive biomarkers of gout. Moreover, these researchers stated that in larger, prospective, cohort studies are needed to validate these findings.

The authors stated that this study had several drawbacks. First, while the direction of effect estimates was relatively consistent, the findings of the meta-analysis exhibited a high degree of heterogeneity, which was inevitable due to different study populations, different body conditions, and different methods of metabolite detection. Second, although these investigators employed a new method of partition functions in converting sample sizes to means and standard deviations, the distribution of the data was still problematic. Third, there was no search of the clinical trial registry database, which may have resulted in an inadequate comprehensive search.

MicroRNAs for the Diagnosis of Gout

Wang et al (2015) stated that microRNAs (miRNAs) are a class of small, non-coding RNAs that function as post-transcriptional repressors of gene expression; they have important roles in many diseases, including inflammatory diseases. Gout is a common arthritis caused by deposition of MSU crystals within joints. Recent studies suggested that miRNAs may be involved in the development of inflammatory arthritis, including acute

gouty arthritis. These investigators discussed relevant publications in order to provide a better understanding on the possible role of miRNAs in gout; miRNAs may act as regulators of gout pathogenesis via several pathways. The authors noted that targeting miRNAs may be a promising strategy in the treatment of gout.

Dalbeth et al (2015) hypothesized that miRNA regulate gene expression of pro-inflammatory cytokines in response to MSU crystals. These researchers stimulated human monocytic THP-1 cells with MSU crystals and examined miRNA and pro-inflammatory cytokine gene expression. The effects of miR-146a over-expression were examined by transfecting THP-1 cells with miR-146a precursor. miR-146a expression was examined in the urate peritonitis model, in peripheral blood mononuclear cells from people with gout and control participants, and in gouty tophus samples. MSU crystals increased miR-146a expression in THP-1 cells, but not other miRNA implicated in iIL-1 β regulation. Over-expression of miR-146a expression reduced MSU crystal-induced IL-1 β , tumor necrosis factor- α (TNF α), monocyte chemoattractant protein-1 (MCP-1) and IL-8 gene expression. In the urate peritonitis model, reduced miR-146a expression was observed during the acute inflammatory response to MSU crystal injection. In people with inter-critical gout, peripheral blood mononuclear cells expressed significantly higher levels of miR-146a, compared with normo-uricemic and hyper-uricemic control participants and those with acute gout flares. Expression of miR-146a was also observed in all tophus samples. The authors concluded that collectively, these data suggested that miR-146a is a transcriptional brake that is lost during the acute inflammatory response to MSU crystals.

Musculoskeletal Ultrasound for Diagnosis of Gout

In a multi-center study, Ogdie and associates (2017) examined the performance of US for the diagnosis of gout using the presence of MSU crystals as the gold standard. These researchers analyzed data from the Study for Updated Gout Classification Criteria (SUGAR), a large, multi-center, observational cross-sectional study of consecutive subjects with at least 1 swollen joint who conceivably may have gout. All subjects underwent arthrocentesis; cases were subjects with confirmed MSU crystals. Rheumatologists or radiologists who were blinded with regard to the results of the MSU crystal analysis performed US on 1 or more

clinically affected joints; US findings of interest were DCS, tophus, and snowstorm appearance. Sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) were calculated. Multi-variable logistic regression models were used to examine factors associated with positive US results among subjects with gout.

Ultrasound was performed in 824 subjects (416 cases and 408 controls). The sensitivity, specificity, PPV, and NPV for the presence of any 1 of the features were 76.9 %, 84.3 %, 83.3 %, and 78.2 %, respectively.

Sensitivity was higher among subjects with a disease duration of greater than or equal to 2 years and among subjects with subcutaneous nodules on examination (suspected tophus). Associations with a positive US finding included suspected clinical tophus (OR 4.77; 95 % CI: 2.23 to 10.21), any abnormality on plain radiography (OR 4.68; 95 % CI: 2.68 to 8.17), and serum urate level (OR 1.31; 95 % CI: 1.06 to 1.62). The authors concluded that US features of MSU crystal deposition had high specificity and high PPV but more limited sensitivity for early gout. The specificity remained high in subjects with early disease and without clinical signs of tophi.

The authors stated that drawbacks of this study included possible selection bias, variation in ultrasonographer training and US machine use, and possible test interpretation bias. Ultrasound was not performed among all subjects in the SUGAR study due to availability of US and trained ultrasonographers at the enrolling sites. However, there were few differences in the subjects who did versus did not undergo US, suggests that there was not significant selection bias in which subjects underwent US. Second, a variety of machines were used and many different ultrasonographers performed the US. Ultrasonographers were mainly rheumatologists who used US in clinical practice although not necessarily certified or radiologists. Although definitions of US features were provided to all ultrasonographers, a standardized scanning protocol was not required. Inter-rater reliability was not assessed; this has been reported and this was not the primary goal of the study. There was some variability in the false-positive and false-negative rates at the individual sites. These researchers did not have the ability to centrally re-read US images. However, this reflected "real world" use of US in clinical practice, increasing the external validity of the results. Understanding US performance in the "real world" was the primary objective of this study.

Third, only clinically affected joints were scanned. Inclusion of additional

asymptomatic joints may have increased the sensitivity. However, the primary goal was to assess the ability of US to assist in diagnosing gout in the symptomatic joint. Test interpretation bias was possible, although should not have a significant influence given that US was performed blinded to synovial fluid analysis. More importantly, ultrasonographers may not have been blinded to all clinical features, for example, it was possible that the presence of visible tophi or other clinically apparent characteristics influenced the interpretation of the US results. However, this also reflected real-life clinical practice, in which ultrasound was used as an additive test to available clinical data. Finally, it was important to recognize that these data were relevant for patients with symptomatic joint swelling. These results could not be applied to diagnosis of gout in patients with asymptomatic hyperuricemia.

Zhang and colleagues (2018) noted that musculoskeletal US is widely used in diagnosing gout, but its accuracy is debatable. These researchers conducted a systematic review and meta-analysis to quantitatively evaluate the value of US in the diagnosis of gout. They systematically searched for publications using Cochrane Library, PubMed/Medline and Embase and manually screened the references of eligible articles for additional relevant publications. Studies were included in this systematic review if they assessed the diagnostic accuracy of US in gout compared to that of the gold standard, demonstration of monosodium urate crystals in joint fluid or tophi. These investigators then conducted quantitative analyses by extracting data from each study and calculating the pooled sensitivity, specificity, positive likelihood ratio (PLR), negative likelihood ratio (NLR) and diagnostic OR (DOR). The summary ROC (sROC) were constructed to obtain the Q^{*}-index and the area under the curve (AUC). A total of 13 studies were included in this meta-analysis. The diagnostic performances of 3 distinctive US features of gout, DCS, the presence of tophi and the snowstorm sign, were evaluated. For person-based evaluations, the pooled sensitivity, specificity, DOR, AUC and Q^{*} were as follows: for the DCS, 66 % (95 % CI: 62 % to 69 %), 92 % (95 % CI: 90 % to 94 %), 25.91 (95 % CI: 11.80 to 56.89), 0.8163 and 0.7503, respectively; for the presence of tophi, 56 % (95 % CI: 52 % to 60 %), 94 % (95 % CI: 92 % to 96 %), 21.11 (95 % CI: 7.84 to 56.89), 0.8928 and 0.8236, respectively; for the snowstorm sign, 31 % (95 % CI: 27 % to 36 %), 91 % (95 % CI: 88 % to 93 %), 4.54 (95 % CI: 3.13 to 6.58), 0.5946 and 0.5712, respectively; and for

simultaneous consideration of these US features, 80 % (95 % CI: 76 % to 83 %), 83 % (95 % CI: 79 % to 86 %), 19.03 (95 % CI: 13.97 to 25.93), 0.889 and 0.8197, respectively. For the joint-/location-based evaluations, the pooled sensitivity, specificity, DOR, AUC and Q* were as follows: for the DCS, 75 % (95 % CI: 68 % to 80 %), 65 % (95 % CI: 59 % to 70 %), 16.90 (95 % CI: 5.10 to 56.03), 0.871 and 0.8014, respectively; and for the presence of tophi, 48 % (95 % CI: 40 % to 57 %), 96 % (95 % CI: 91 % to 99 %), 30.20 (95 % CI: 9.23 to 98.87), 0.8776 and 0.8081, respectively. The authors concluded that in this meta-analysis, relatively high specificity but modest or low sensitivity were demonstrated in the diagnosis of gout using each of the 3 US features for person-based evaluations. Simultaneous consideration of these US findings may improve the diagnostic sensitivity. However, the double contour sign alone was weak in the differentiation of gout and non-gout for joint-/location-based evaluations. They stated that further well-designed studies are still needed to support the current findings.

The authors stated that this study had several drawbacks. First, although 13 studies were included in the meta-analysis, sub-group analyses were conducted based on a small number of studies. Second, these researchers included both cross-sectional studies and case-control studies. The quality of the primary studies greatly influenced the quality of the meta-analysis. Some cross-sectional studies enrolled patients with inflamed joints without clear diagnoses, which represented those mostly likely to accept US examinations in clinical practice and gain benefits; however, other studies were conducted in established (somewhat advanced) gout patients in rheumatology clinics where the diagnosis was clear. In addition, the definitions of the control groups were different among the case-control studies included in this meta-analysis; some studies included control patients with inactive joint diseases that were unlikely to be gout, while others included healthy participants. Thus, selection bias was inevitable. Third, the qualification of the sonographers, the device used, duration of symptoms, the US features taken into overall consideration, interpretation of US images among sonographers, the number of examined joints in person-based evaluations and other methodological characteristics varied across studies. These investigators stated that future studies are needed to refine the study design and examine the performance of US at specific sites and at specific time-points in the disease course of gout.

Furthermore, follow-up should be recommended to observe the longitudinal changes of US features along and their relationship with serum urate acid levels.

In a prospective study, Strobl and associates (2018) compared findings of US with DECT findings in patients presenting with suspected gouty knee arthritis. This trial included 65 patients (52 men and 13 women; median age of 61.7 years [range of 38 to 87 years]) with an initial clinical diagnosis of acute gouty knee arthritis who underwent DECT performed using a 128-MDCT scanner and US performed using a 5-18-MHz transducer. Both intra- and extra-articular findings obtained using each modality were tabulated. DECT identified gout as the final diagnosis for 52 of 65 patients (80.0 %). An alternative diagnosis was confirmed for the remaining 13 patients; US detected gout in 31 of 52 patients (sensitivity, 59.6 %) and produced findings negative for gout in 7 of 13 patients (specificity, 53.8 %). The DCS on US was positive for gout in 23 of 52 patients (44.2 %) and negative in 12 of 13 patients (92.3 %). Extra-articular urate deposition was identified by DECT in 44 of 52 patients, compared with identification by US in 11 of 52 patients ($p < 0.001$). The authors concluded that the sensitivity of US for the diagnosis of gouty knee arthritis was limited, particularly with respect to extra-articular urate deposition. The DCS was the single most valuable sign for the assessment of gouty knee arthritis by US.

In a prospective study, Klauser and co-workers (2018) compared findings of US with DECT in patients presenting with suspected gouty hand and wrist arthritis. This trial included 180 patients (136 men and 44 women, age range of 31 to 94 years; mean age of 65.9 years) with an initial clinical diagnosis of acute gouty arthritis who underwent DECT and US examination. Intra- and extra-articular findings of each modality were tabulated and calculated with DECT as gold standard. The final diagnosis of gout was positive in 97/180 patients (53.9 %) by DECT, an alternative diagnosis confirmed in 83 patients; US showed a sensitivity of 70.1 % (extra-articular: 42.5 %, $p < 0.0001$; intra-articular: 80.3 %, $p = 0.14$) and specificity of 51 %. The DCS was present in 58/61 patients with a positive US study for intra-articular gout (95.1 %). The authors concluded that the sensitivity of US for diagnosis of gouty arthritis in hand and wrist was limited, particularly with respect to extra-articular urate deposition. The DCS was the most sensitive sign for the assessment of

gouty hand and wrist arthritis by US.

The Centers for Disease Control and Prevention's webpage on "Gout" (CDC, last updated April 3, 2018) states that "A medical doctor diagnoses gout by assessing your symptoms and the results of your physical examination, X-rays, and lab tests. Gout can only be diagnosed during a flare when a joint is hot, swollen, and painful and when a lab test finds uric acid crystals in the affected joint".

Nanoparticles for the Treatment of Gouty Arthritis

In a systematic review and meta-analysis, Zhu et al (2023) examined the effects of nanoparticles on gouty arthritis, and provided evidence for the pre-clinical use of nanoparticles in gouty arthritis and ideas for nanomedicine improvement for nanoparticle researchers. These investigators searched 5 databases including the Cochrane Library, PubMed, Scopus, Web of Science, and Embase for eligible studies until April 2022. The quality of the selected studies was examined by SYRCLE's risk of bias (RoB) tool, and the random-effects model was used to calculate the overall effect sizes of weighted mean differences (WMD). A total of 10 studies met the inclusion criteria. Results showed that nanoparticles were effective in reducing UA levels (WMD: -4.91; 95 % CI: - 5.41 to - 4.41; $p < 0.001$), but were not better than allopurinol (WMD: -0.20; 95 % CI: - 0.42 to 0.02; $p = 0.099$). It was worth noting that the nanoparticles were safer than allopurinol. Subgroup analyses indicated that nanoparticle encapsulated substance, animal species, nanoparticle dosage, animal quantity, and animal gender were all sources of heterogeneity. The authors concluded that the findings of this systematic review and meta-analysis showed that nanoparticles could effectively reduce the level of UA in animal models of gout. These researchers stated that nanoparticles might become effective medications in the treatment of gouty arthritis; however, the results are inconclusive. They stated that larger, well-designed studies with longer follow-up are needed to show that nanoparticles can be used in pre-clinical treatment of gouty arthritis.

The authors stated that this systematic review and meta-analysis had several drawbacks. First, the number of included studies and sample sizes were small. Second, the included studies were only published in

English, which could be incomprehensive. Third, the heterogeneity of the meta-analysis was high, but this was unavoidable because the included studies varied in species, quantity, and gender of animals, dosage, and nanoparticle encapsulated substances; thus, subgroup analyses and sensitivity analyses were used to examine possible sources of heterogeneity. Fourth, the funnel plots of nanoparticles compared with the model group and allopurinol were asymmetric, and Egger's test also suggested the possibility of publication bias. This might be related to the selective reporting of studies, with only 4 of the studies included in the review being able to extract data for meta-analysis. Meanwhile, to a certain extent, it showed that the research in this area was still in the development stage and had great application prospects. In the future, more high-quality literature is needed to support the research ideas. These investigators stated that the findings of this meta-analysis would provide a reference for the further development of nanomedicine.

Next-Generation Sequencing Profiling of Mitochondrial Genomes for Diagnosis of Gout

Tseng and colleagues (2018) noted that accumulating evidence implicates mitochondrial DNA (mtDNA) alleles, which are independent of the nuclear genome, in disease, especially in human metabolic diseases. However, this area of investigation has lagged behind in researching the nuclear alleles in complex traits, for example, in gout. In this exploratory study, next-generation sequencing (NGS) was utilized to examine the relationship between mtDNA alleles and phenotypic variations in 52 men (age of 51.60 ± 10.82 years) with gout and 104 age-matched men non-gout controls (age of 51.61 ± 10.78 years) from the Taiwan Biobank whole-genome sequencing samples. Differences from a reference sequence (GRCh38) were identified. The sequence kernel association test (SKAT) was applied to identify gout-associated alleles in mitochondrial genes. The tools Polymorphism Phenotyping, Sorting Intolerant From Tolerant (SIFT), Predict the pathology of Mutations (PMUT), Human Mitochondrial Genome Database (mtDB), Multiple Alignment using Fast Fourier Transform (MAFFT), and Mammalian Mitochondrial tRNA Genes (Mamit-tRNA) were used to evaluate pathogenicity of alleles. Validation of selected alleles by quantitative polymerase chain reaction of single nucleotide polymorphisms (qPCR SNPs) was also performed. These investigators identified 456 alleles in

patients with gout and 640 alleles in non-gout controls with 274 alleles shared by both. Mitochondrial genes were associated with gout, with MT-CO3, MT-TA, MT-TC, and MT-TT containing potentially pathogenic gout-associated alleles and displaying evidence of gene-gene interactions. All heteroplasmy levels of potentially pathogenic alleles exceeded metabolic thresholds for pathogenicity. Validation assays confirmed the NGS results of selected alleles. Among them, potentially pathogenic MT-CO3 alleles correlated with high-density lipoprotein (HDL) levels ($p = 0.034$). The authors concluded that the findings of this exploratory study suggested that mitochondrial alleles potentially play a role in the pathogenesis of gout and identify patient subgroups with distinct clinical phenotypes. Further validation and functional studies to clarify underlying mechanisms are recommended.

Pegloticase (Krystexxa)

U.S. Food and Drug Administration (FDA)-Approved Indications

- Krystexxa is indicated for the treatment of chronic gout in adult patients refractory to conventional therapy.
- Limitations of Use: Krystexxa is not recommended for the treatment of asymptomatic hyperuricemia.

Pegloticase, branded as Krystexxa (Horizon Therapeutics USA, Inc), is a genetically engineered PEGylated recombinant porcine uricase (urate oxidase). Pegloticase metabolizes uric acid into soluble allantoin for excretion by the kidney with hydrogen peroxide and carbon dioxide as oxidative byproducts.

On September 14, 2010, the FDA approved pegloticase (Krystexxa) for the treatment of chronic gout in adults who are refractory to or can not tolerate conventional therapy. Patients who have failed to normalize serum uric acid (to less than 6 mg/dL) with xanthine oxidase inhibitors at the maximum medically appropriate dose for at least 3 months are deemed refractory. The maximum recommended dosages of allopurinol [Zyloprim] and febuxostat [Uloric] for gout are 800 mg/day and 80 mg/day, respectively. The approval was based on 2 replicate, multi-center, randomized, double-blind, placebo-controlled clinical studies of 6 months duration (a total of 212 patients). Patients were randomized to receive

pegloticase every 2 weeks or every 4 weeks or placebo in a 2:2:1 ratio. The primary endpoint in both trials was the proportion of patients who achieved PUA less than 6 mg/dL for at least 80 % of the time during month 3 and month 6. The data in both clinical studies demonstrated that a greater proportion of patients treated with pegloticase every 2 weeks achieved urate lowering to below 6 mg/dL than patients receiving placebo. During the first 6 months of treatment, 47 % ($p < 0.001$) and 38 % ($p < 0.001$) of patients in the pegloticase arms of the 2 clinical studies achieved the primary efficacy endpoint, compared with 0 % of patients in the placebo arm.

The effect of treatment with pegloticase on tophi was a secondary efficacy endpoint of the clinical studies and was assessed using standardized digital photography, image analysis and a central reader blinded to treatment assignment. Baseline tophi was found in 71 % of patients. A pooled analysis of data from both clinical studies at month 6 demonstrated that 45 % ($p < 0.02$) of patients with tophi treated with pegloticase every 2 weeks achieved a complete response, defined as 100 % resolution of at least one target tophus, no new tophus appearing and no single tophus showing progression, compared to 8 % of patients receiving placebo.

Since 25% of patients in the clinical trials experienced a severe allergic reaction when receiving an infusion of Krystexxa, health care providers should dispense an anti-histamine and a corticosteroid to their patients beforehand to minimize the risk of such a reaction. Other reactions included chest pain, constipation, gout flare, injection site bruising, irritation of the nasal passages, nausea and vomiting. The drug is administered to patients every 2 weeks as an intravenous infusion; it should not be administered as an intravenous push or bolus.

Pegloticase is contraindicated in patients with glucose-6-phosphate dehydrogenase (G6PD) deficiency due to the risk of methemoglobinemia and hemolysis. It is recommended that patients at higher risk for G6PD deficiency (e.g., patients of African or Mediterranean ancestry) be screened for G6PD deficiency before starting pegloticase.

The labeling for Krystexxa includes a boxed warning for anaphylaxis, infusion reactions, G6PD deficiency-associated hemolysis, and

methemoglobinemia. Anaphylaxis and infusion reactions can occur during or after the administration of Krystexxa, with anaphylaxis potentially happening with any infusion, including the first, typically within two hours, although delayed hypersensitivity reactions have also been noted. Therefore, Krystexxa should be administered in healthcare settings by providers equipped to manage these reactions. Pre-medication with antihistamines and corticosteroids is recommended, along with close monitoring for anaphylaxis for an appropriate duration post-administration. Additionally, serum uric acid levels should be monitored before each infusion, and treatment should be discontinued if levels exceed 6 mg/dL, especially if two consecutive readings are above this threshold. Patients at risk for G6PD deficiency should be screened prior to starting Krystexxa, as hemolysis and methemoglobinemia have been reported in these individuals, and Krystexxa is contraindicated in patients with G6PD deficiency.

In a 52-week controlled trial comparing Krystexxa administered with methotrexate to Krystexxa alone, participants were given standardized prophylaxis for infusion reactions and were removed from treatment if their serum uric acid levels surpassed 6 mg/dL at two consecutive visits to reduce the risk of anaphylaxis. One subject in the Krystexxa and methotrexate group (1%) experienced anaphylaxis during the first infusion, while no cases occurred in the Krystexxa-only group. In pre-marketing clinical trials with Krystexxa alone, the drug was not discontinued for elevated uric acid levels, and anaphylaxis was reported in 6.5% (8/123) of subjects on a biweekly regimen and 4.8% (6/126) on a monthly regimen, with no cases in the placebo group. Anaphylaxis typically occurred within two hours of treatment, with diagnostic criteria including skin or mucosal involvement and either airway compromise or reduced blood pressure, often accompanied by symptoms such as wheezing or edema. Notably, some cases occurred in patients pre-treated with antihistamines, corticosteroids, or acetaminophen, suggesting that the reported frequency may underestimate the true incidence. Krystexxa should be administered in a healthcare setting by providers prepared to manage anaphylaxis, with patients pre-treated with antihistamines and corticosteroids. Anaphylaxis can occur with any infusion, including the first, and patients should be closely monitored for a suitable period post-administration. The risk of anaphylaxis increases with uric acid levels above 6 mg/dL, especially with two consecutive

elevated readings, necessitating monitoring of serum uric acid levels before infusions and discontinuation of treatment if levels rise above this threshold. To avoid potential interference with uric acid levels, it is recommended that patients discontinue oral urate-lowering medications before starting Krystexxa and refrain from initiating such therapy during treatment.

Patients treated with Krystexxa may experience infusion reactions, so it is essential to pre-medicate and monitor them during treatment. Those at risk for G6PD deficiency should be screened, and Krystexxa should not be given to patients with this deficiency. Additionally, gout flare prophylaxis is advised for at least the first six months of therapy with Krystexxa. It is also important to note that there is a potential for exacerbation of congestive heart failure in some patients.

Immunogenicity - Anti-pegloticase antibodies developed in 92% of patients treated with Krystexxa every 2 weeks, and 28% for placebo. High anti-pegloticase antibody titer was associated with a failure to maintain pegloticase-induced normalization of uric acid. There was a higher incidence of infusion reactions in patients with high anti-pegloticase antibody titer: 53% (16 of 30) in the Krystexxa every 2 weeks group compared to 6% in patients who had undetectable or low antibody titers.

No controlled trial data are available on the safety and efficacy of re-treatment with Krystexxa after stopping treatment for longer than 4 weeks. Due to the immunogenicity of Krystexxa, patients receiving re-treatment may be at increased risk of anaphylaxis and infusion reactions. Therefore, patients receiving pre-treatment after a drug-free interval should be monitored carefully.

The safety and effectiveness of Krystexxa in pediatric patients less than 18 years of age have not been established.

The most common adverse reactions of Krystexxa co-administered with methotrexate (5% or more of patients) include gout flares, arthralgia, COVID-19, nausea and fatigue. The most common adverse reaction of Krystexxa alone (5% or more of patients) include gout flares, infusion reactions, nausea, contusion or ecchymosis, nasopharyngitis,

constipation, chest pain, anaphylaxis and vomiting.

Plasma Profiling of Amino Acids for Differential Diagnosis of Acute Gout from Asymptomatic Hyperuricemia

Luo and colleagues (2018) stated that gout and hyperuricemia are highly prevalent metabolic diseases caused by high level of uric acid. Amino acids (AAs) involve in various biochemical processes including the biosynthesis of uric acid. However, the role of AAs in discriminating gout from hyperuricemia remains unknown. These investigators reported that the plasma AAs profile can distinguish acute gout (AG) from asymptomatic hyperuricemia (AHU). They established a liquid chromatography-mass spectrometry (LC-MS)/MS-based method to measure the plasma AAs without derivatization for the AG and AHU patients, and healthy controls. These researchers found that the plasma profiling of AAs separated the AG patients from AHU patients and controls visually in both principal component analysis and orthogonal partial least-squares discriminant analysis (OPLS-DA) models. In addition, L-isoleucine, L-lysine, and L-alanine were suggested as the key mediators to distinguish the AG patients from AHU and control groups based on the S-plot analysis and variable importance in the projection values in the OPLS-DA models, volcano plot, and the ROC. In addition, the saturation of monosodium urate in the AA solutions at physiologically mimic status supported the changes in plasma AAs facilitating the precipitation of monosodium urate. The authors concluded that the findings of this study suggested that L-isoleucine, L-lysine, and L-alanine could be the potential markers to distinguish the AG from AHU when the patients had similar blood levels of uric acid, providing new strategies for the prevention, treatment, and management of acute gout.

Pricking-Blood Therapy for the Treatment of Acute Gouty Arthritis

Li and colleagues (2020) stated that acute gouty arthritis (AGA) is a joint inflammatory reaction that affects the daily quality of patients. Previous reviews of pricking-blood therapy (a kind of external therapy of traditional Chinese medicine) for AGA have been growing, but a systematic review is not available. These researchers will systematically examine the safety and effectiveness of pricking-blood therapy in the treatment of

patients with AGA. They will search for relevant literature through Chinese and English databases, with the retrieval deadline being December 2020. Databases include PubMed, Embase, Web of Science, the Cochrane Library, China National Knowledge Infrastructure, the Chongqing VIP Chinese Science and Technology Periodical Database, Wanfang Database, and China Biomedical Literature Database. These investigators will also manually search Chinese Acupuncture & Moxibustion, Acupuncture Research, Chinese Clinical Trial Register, and unpublished studies or references. According to the inclusion and exclusion criteria, the literature will be screened, and the data are extracted independently by the 2 researchers; the primary outcomes are the total effective rate and VAS score. RevMan 5.3.5. software will be used for statistical analysis. According to the GRADE approach, each evidence of outcome quality will be appraised. The authors hope that this study will provide a comprehensive review of current evidence for the safety and effectiveness of pricking-blood therapy in the treatment of patients with AGA and guide clinical decision-making.

Serum Ionized Magnesium and Gouty Arthritis

Li et al (2024) stated that gouty arthritis is a common inflammatory arthritis. The recurrent gout attacks severely damage the joint's function; resulting in bone erosion as well as affecting bone metabolism. The role of magnesium (Mg) ions in bone homeostasis has been recognized, whereas its specific relationship with gouty bone erosion remains unclear.

In a cross-sectional study, these researchers examined the association between serum ionized Mg levels and bone erosion in patients with gout arthritis. A total of 769 patients with gout arthritis were included in this trial. Subjects were classified into 4 groups based on the quartiles of the serum ionized Mg level. Logistic regression analysis examined the association between serum ionized Mg and bone erosion. Compared to patients without bone erosion, serum ionized Mg levels were lower in gout patients with bone erosion ($p < 0.001$). When dividing serum ionized Mg into quartiles, the prevalence rate of bone erosion in group Q1, representing the patients with the lowest serum ionized Mg levels, was notably higher than in Q2, Q3, and Q4 (60.2 % versus 43.6 %, 45.6 %, 40.3 %, $p < 0.001$). Multiple logistic regression analysis showed that patients in Q2-Q4 had a lower OR of bone erosion compared to those in Q1 (ORs were 0.520, 0.533, and 0.411 in Q2-Q4, respectively, $p <$

0.001). The authors concluded that the incidence of bone erosion was higher in gout arthritis patients with lower serum ionized Mg levels. High serum ionized Mg levels may be an independent protective factor for bone erosion in gout arthritis; therefore, Mg supplementation may be a promising approach to prevent or slow down the development of bone erosion in gouty arthritis. Moreover, these investigators stated that further research is needed to confirm the safety and effectiveness of Mg supplements in the treatment of gout.

The authors stated that this study had several drawbacks. First, this was a cross-sectional study, which determined association, not causality; thus, prospective, interventional studies are still needed to evaluate the effect of Mg on bone erosion. Second, serum ionized Mg levels are closely associated with dietary habits; however, further analysis of dietary patterns could not be carried out due to a lack of dietary data. Third, the study population was over 95% male; therefore, whether these findings could be generalized to women is uncertain.

Sodium-Glucose Transport Protein 2 (SGLT2) Inhibitors for Prevention of Incident Gout in Patients with Type 2 Diabetes Mellitus

Chung et al (2021) noted that the use of sodium-glucose transport protein 2 inhibitors (SGLT2i) is currently a standard intervention in patients with type 2 diabetes mellitus (T2DM) and exerts favorable pleiotropic effects to consistently lower blood urate levels. However, currently, no association between SGLT2i use and the incidence of gout have been established. In a cohort study, these investigators examined if prescribed SGLT2i are associated with lower gout incidence in patients with T2DM. All patients with incident T2DM in Taiwan National Health Institution databases between May 1, 2016, and December 31, 2018, were retrospectively analyzed. As a comparator, patients using dipeptidyl peptidase 4 inhibitor (DPP4i) were included. A total of 47,905 individuals receiving an SGLT2i and 183,303 receiving a DPP4i were evaluated, along with 47,405 pairs of patients using an SGLT2i or DPP4i in 1:1 propensity score-matched analyses. Data analysis was performed from April 1 to June 30, 2021. A gout diagnosis was based on the International Classification of Diseases, 9th Revision, Clinical Modification (ICD-9-CM) and the International Statistical Classification of

Diseases, 10th Revision, Clinical Modification (ICD-10-CM). Multiple Cox proportional hazards regression models were used to calculate hazard ratios (HRs) and 95 % CIs. A total of 231,208 patients with T2DM were included in the population; 113,812 individuals (49.22 %) were women, and the mean (SD) age was 61.53 (12.86) years. The overall gout incidence was 20.26 per 1,000 patient-years for SGLT2i users and 24.30 per 1,000 patient-years for DPP4i users. When potential risk factors were adjusted in the propensity score-matched population, use of SGLT2i was associated with a lower risk of gout (HR, 0.89; 95 % CI: 0.82 to 0.96) compared with DPP4i, especially for patients receiving dapagliflozin (HR, 0.86; 95 % CI: 0.78 to 0.95). A sensitivity analysis, carried out when a gout diagnosis was ascertained using the ICD-9-CM or ICD-10-CM code with gout-related medication, also showed a significantly lower risk for gout incidence of 15 % with SGLT2i (HR, 0.85; 95 % CI: 0.74 to 0.97). Subgroup analysis indicated that SGLT2i benefits in patients with T2DM to achieve a lower gout risk were not different across subgroups. The authors concluded that the findings of this study suggested that patients with T2DM who were receiving SGLT2i may have a lower risk for gout compared with those receiving DPP4i. Moreover, these researchers stated that further studies with longer follow-up durations should be implemented.

The authors stated that this study had several drawbacks. First, detailed laboratory values (e.g., blood urate levels) were not included in the National Health Insurance database, potentially influencing study outcomes. Second, in Taiwan, SGLT2i were introduced in 2016; thus, the sample size and event rate were decreased in the subgroup analysis, which would have lessened the power to detect differences. Third, observation and outcome periods were relatively short. However, a significant association of a lower risk of gout with SGLT2i was still observed. These investigators expected that this incidence difference to grow with longer follow-up periods because reductions in blood urate levels should translate to lower incident gout risks over time. Fourth, gout ICD-9-CM and ICD-10-CM codes were not validated in the database. Furthermore, a previous validation study was limited by the uncertainty of external validity criteria in the general population. Nevertheless, a gout diagnosis was strengthened by use of ICD-9-CM and ICD-10-CM codes in combination with medication information, which also noted an association between SGLT2i and a lower risk of gout.

In a systematic review and meta-analysis, Banerjee et al (2022) examined the effect of SGLT2i on adverse gout events in patients with T2DM. These investigators searched PubMed/Medline, Embase, and Web of Science databases using appropriate keywords/MeSH/Emtree terms till January 25, 2022, to identify observational studies, RCTs or post-hoc analysis reporting incident gout events and/or commencement of anti-gout drug in patient with T2DM receiving SGLT2i versus those not receiving SGLT2i. Subgroup analyses were carried out using comparators as placebo/other anti-diabetic drugs and presence/absence of baseline hyperuricemia (uric acid 7 or higher, or less than 7 mg/dL); HR with 95 % CI were calculated. These researchers identified 5 studies (3 observational, 2 post-hoc analysis of RCTs) pooling data retrieved from 568,010 patients with T2DM. Pooled analysis showed that SGLT2i use was associated with 30 % reduction in incident gout events/gout flares (HR 0.70, 95 % CI: 0.59 to 0.84, $p < 0.001$, $I^2 = 84\%$). Sensitivity analysis after excluding the retrospective observational study showed similar estimates (HR 0.65, 95 % CI: 0.60 to 0.70, $p < 0.001$, $I^2 = 0\%$). Subgroup analysis of data retrieved only from RCTs also showed significant benefits (HR 0.74, 95 % CI: 0.55 to 0.98, $p = 0.03$, $I^2 = 0\%$). Pooled analysis of data from 2 studies showed that SGLT2i use resulted in a significant reduction in the need for commencement of new anti-gout drug (pooled HR 0.58, 95 % CI: 0.4 to , 0.71, $p < 0.001$, $I^2 = 0\%$). Consistent benefits were also observed for subgroup without baseline hyperuricemia (pooled HR 0.65, 95 % CI: 0.47 to 0.89, $p < 0.01$, $I^2 = 0\%$). The authors concluded that SGLT2i may potentially prevent gout-related AEs in patients with T2DM.

Appendix

Appendix A

Clinical reasons for not completing a three-month trial with allopurinol, febuxostat, and probenecid (examples, not all inclusive):

- Member experienced a severe allergic reaction to the medication
- Member experienced toxicity with the medication

- Member could not tolerate the medication
- Member's current medication regimen has a significant drug interaction
- Member has severe renal dysfunction (allopurinol)
- Member has known blood dyscrasias or uric acid kidney stones (probenecid)
- Member has renal insufficiency (i.e., glomerular filtration rate 30 mL/minute or less) (probenecid)
- Member has end stage renal impairment (febuxostat)
- Member has a history of CVD or a new CV event (febuxostat).

Appendix B

Contraindications / clinical reasons to avoid oral methotrexate therapy (examples, not all inclusive):

- Clinical diagnosis of alcohol use disorder, alcoholic liver disease, or other chronic liver disease
- Breastfeeding
- Blood dyscrasias (e.g., thrombocytopenia, leukopenia, significant anemia)
- Elevated liver transaminases
- History of intolerance or adverse event
- Hypersensitivity
- Interstitial pneumonitis or clinically significant pulmonary fibrosis
- Myelodysplasia
- Pregnancy or currently planning pregnancy
- Renal impairment
- Significant drug interaction.

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